

Ethernet Theory of Operation

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INTRODUCTION

This document specifies the theory and operation of the Ethernet technology found in PIC[®] MCUs with integrated Ethernet and in stand-alone Ethernet controllers.

Ethernet technology contains acronyms and terms defined in Table 1.

APPLICATIONS

Ethernet is an asynchronous Carrier Sense Multiple Access with Collision Detect (CSMA/CD) protocol/interface, with a payload size of 46-1500 octets. With data rates of tens to hundreds of megabits/second, it is generally not well suited for low-power applications.

However, with ubiquitous deployment, internet connectivity, high data rates and limitless range expansibility, Ethernet can accommodate nearly all wired communications requirements. Potential applications include:

- Remote sensing and monitoring
- Remote command, control and firmware updating
- Bulk data transfer
- Live streaming audio, video and media
- Public data acquisition (date/time, stock quotes, news releases, etc.)

THEORY OF OPERATION

Ethernet is a data link and physical layer protocol defined by the IEEE 802.3[™] specification. It comes in many flavors, defined by maximum bit rate, mode of transmission and physical transmission medium.

- Maximum Bit Rate (Mbps/s): 10, 100, 1000, etc.
- Mode of Transmission: Broadband, Baseband
- Physical Transmission Medium: Coax, Fiber, UTP, etc.

TABLE 1: ETHERNET GLOSSARY

Term	Definition
CRC	Cyclic Redundancy Check: Type of checksum algorithm used when computing the FCS for all Ethernet frames and the hash table key for hash table filtering of receive packets.
DA	Destination Address: The 6-octet destination address field of an Ethernet frame.
ESD	End-of-Stream Delimiter: In 100 Mb/s operation, the ESD is transmitted after the FCS (during the inter-frame gap) to denote the end of the frame.
FCS	Frame Check Sequence: The 4-octet field at the end of an Ethernet frame that holds the error detection checksum for that frame.
IP	Internet Protocol: Refers either to IPv4 or IPv6.
LAN	Local Area Network or Large Area Network.
MAC	Media Access Control: The block responsible for implementing the Media Access Control functions of the Ethernet specification.
MAC Address	A 6-octet number representing the physical address of the node(s) on an Ethernet network. Every Ethernet frame contains both a source and destination address, both of which are MAC addresses.
MDI	Medium Dependent Interface or Management Data Input.
MDO	Management Data Output.
MDIO	Management Data Input/Output.
MII	Media Independent Interface: Standard 4-bit interface between the MAC and the PHY for communicating TX and RX frame data. In 10 Mb/s mode, the MII runs at 2.5 MHz; in 100 Mb/s mode, it runs at 25 MHz.
MIIM	MII Management: Set of MII sideband signals used for accessing the PHY registers.

以太网工作原理

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简介

本文档规定了 PICfi 微控制器中集成的以太网以及独立以太网控制器中以太网技术的理论与操作。

以太网技术包含表 1 中定义的缩写和术语。

应用程序

以太网是一种异步载波监听多路访问 with 碰撞检测 (载波监听多路访问/冲突检测协议) 协议/接口，有效载荷大小为 46-1500 字节。以每秒数十至数百兆比特的数据速率，它通常不适合低功耗应用。

然而，凭借广泛部署、互联网连接、高速数据速率以及无限的可扩展性，以太网能够满足几乎所有有线通信需求。潜在应用包括：

- 遥感与监测
- 远程命令、控制及固件更新
- 批量数据传输
- 直播音视频及媒体
- 公共数据采集（日期/时间、股票行情、新闻发布等）

工作原理

以太网是由 IEEE 802.3[™] 规范定义的链路层和物理层协议。它有多种形式，由最大比特率、模式等定义。

传输和物理传输介质。

- 最大比特率 (Mbps/s): 10, 100, 1000 等。
- 传输模式：宽带，基带
- 物理传输介质：同轴电缆、光纤、UTP 等。

表 1：以太网术语表

Term	定义
CRC	循环冗余校验：计算所有以太网帧校验序列（FCS）时使用的校验和算法类型，以及用于接收数据包哈希表过滤的哈希表键类型。
DA	目的地址：以太网帧中的 6 字节目的地址字段。
ESD	流结束符 (ESD)：在 100 Mb/s 操作模式下，ESD 在帧校验序列 (FCS) 之后传输（位于帧间间隙），以表示帧的结束。
FCS	帧校验序列：以太网帧末尾的 4 字节字段，用于存储该帧的错误检测校验和。
IP	互联网协议：指 IPv4 或 IPv6 协议族。
LAN	局域网或广域网。
MAC	媒体访问控制：负责实现以太网规范中媒体访问控制功能的模块。
MAC 地址	表示以太网网络上节点物理地址的 6 字节数字。每个以太网帧都包含源地址和目的地址，两者均为 MAC 地址。
MDI	中端依赖接口或管理数据输入。
MDO	管理数据输出。
MDIO	管理数据输入/输出。
MII	媒体独立接口：MAC 与 PHY 之间用于通信发送和接收帧数据的标准 4 位接口。在 10 Mb/s 模式下，MII 运行频率为 2.5 MHz；在 100 Mb/s 模式下，运行频率为 25 MHz。
MIIM	MII 管理：用于访问 PHY 寄存器的 MII 侧带信号组。

TABLE 1: ETHERNET GLOSSARY (CONTINUED)

Term	Definition
OUI	Organizationally Unique Identifier: The upper three octets of a MAC address are referred to as the OUI, and typically are assigned to an organization or company. Microchip's OUI is 00-04-A3h.
Octet	In Ethernet terms, one 8-bit byte.
Packet Buffer	The physical or virtual memory where all transmit and receive packets (frames) are stored.
PHY	The block that implements the Ethernet physical layer.
RAM	Random Access Memory (normally volatile memory).
Receive Buffer	Logical portion of the packet buffer used to store received packets.
RX	Receive.
SA	Source Address: The 6-octet source address field of an Ethernet frame.
SFD	Start Frame Delimiter: The single octet field of an Ethernet frame that marks the start of a frame.
SPI	Serial Peripheral Interface.
SSD	Start-of-Stream Delimiter: In 100 Mb/s Ethernet, the first octet of the preamble is known as the SSD and is encoded differently from the rest of the preamble.
Station Address	The Station Address is the MAC address of the Ethernet node. It is typically compared against the destination address in a received Ethernet frame to determine if the frame should be received or not. On the transmit side, it is typically transmitted as the source address of an Ethernet frame.
Transmit Buffer	Logical portion of the packet buffer used to store packets to be transmitted.
TX	Transmit.
RMII	Reduced Media Independent Interface: A 2-bit version of the MII.
SMII	Serial Media Independent Interface: A 1-bit version of the MII.
NRZI	Non-Return-to-Zero Inverted: A binary code in which a logical one is represented by a signal transition and a logical zero is represented by the lack of a transition.

表 1：以太网术语表（续）

Term	定义
OUI	组织唯一标识符：MAC 地址的高三位字节被称为 OUI，通常分配给一个组织或公司。Microchip' 的 OUI 为 00-04-A3h。
字节	在以太网术语中，一个 8 位字节。
数据包缓冲区	存储所有发送和接收数据包（帧）的物理或虚拟内存。
PHY	实现以太网物理层的块。
RAM	随机存取存储器（通常为易失性存储器）。
接收缓冲区	用于存储接收到的数据包的逻辑数据包缓冲区部分。
RX	接收。
SA	源地址：以太网帧中的 6 字节源地址字段。
SFD	帧起始符：以太网帧中用于标记帧起始的单个字节字段。
SPI	串行外设接口。
SSD	起始流分隔符：在 100 Mb/s 以太网中，前导码的第一个字节被称为 SSD，其编码方式与前导码的其余部分不同。
站点地址	站点地址是以太网节点的 MAC 地址。它通常与接收到的以太网帧中的目的地址进行比较，以确定是否接收该帧。在发送端，它通常作为以太网帧的源地址进行传输。
传输缓冲区	用于存储待传输数据包的逻辑数据包缓冲区部分。
TX	发送。
RMII	降低的媒体独立接口：MII 的 2 位版本。
SMII	串行媒体独立接口：MII 的 1 位版本。
NRZI	非归零反相：一种二进制编码，其中逻辑"1"由信号跳变表示，逻辑"0"由无跳变表示。

PROTOCOL STACK

The easiest way to understand the role that Ethernet plays is by looking at a protocol stack, which describes a complete protocol or set of protocols in a layered approach (see Figure 1).

Frame/Package Encapsulation

To understand how Ethernet works, it is first necessary to understand the concept of packet encapsulation, and how the protocol stack fits into this concept.

Each layer of the protocol stack is responsible for a particular level of functionality. As an example, the physical layer is concerned with the actual electrical transmission of bits across a medium. Each higher layer in the model utilizes the underlying layers in a somewhat independent fashion (meaning little or no overlap in functions between the layers).

This layered approach is implemented through the use of encapsulation. This concept can best be explained using the example shown in Figure 2. This example shows how each layer associated with a web browser session maps to the protocol stack model.

Starting at the application layer, the web browser would generate an HTTP request using an application-specific command. This request would then be passed down to the TCP layer, which would construct a TCP packet consisting of a TCP header and TCP data. The TCP header contains information particular to the TCP protocol, such as packet sequencing information, checksum information and the source and destination port number (HTTP typically has a port number of 80).

At the IP protocol level, an IP datagram is constructed to hold the TCP packet. Similar to the TCP packet, the IP datagram consists of an IP header and IP data. The IP header contains information such as the type of service, checksum information, protocol type (06h for TCP), and the source and destination IP addresses. The data field of the IP datagram contains the complete TCP packet to be transmitted.

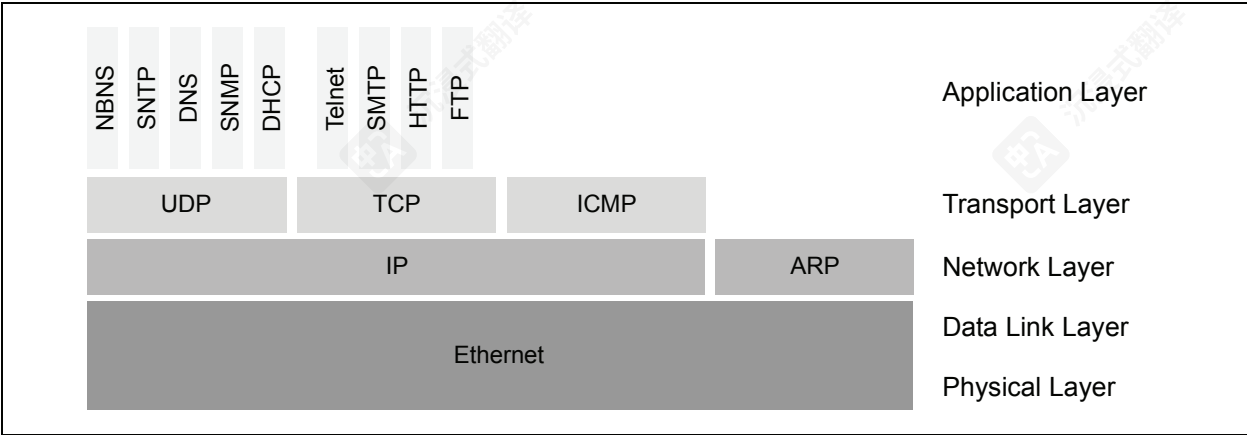
At the data link/physical layer, the IP datagram is transported across the network using the IEEE 802.3 protocol. A MAC (IEEE 802.3) frame consists of a MAC header and a MAC payload (data). The MAC header contains information about the MAC frame, such as the source MAC address, the destination MAC address and the length of the frame. The payload field contains the complete IP datagram to be transported.

Note that the various addresses encapsulated within each protocol are different, and typically, have no fixed relationship to one another. In our example, the TCP packet uses a port number, which is typically assigned based on the application layer protocol (i.e., port 80 for HTTP). The IP datagram uses an IP address, which is statically or dynamically assigned out of a pool of available internet addresses, and the MAC frame uses MAC addresses, which are assigned to the particular piece of hardware.

Note 1: The terms “MAC frame”, “Ethernet frame” and “IEEE 802.3 frame” are used interchangeably in this document.

2: The terms “packet”, “frame” and “datagram” are often used interchangeably. These terms apply to specific protocols, such as an IEEE 802.3 frame, a TCP packet or an IP datagram.

FIGURE 1: INTERNET PROTOCOL STACK



协议栈

理解以太网作用的最简单方法是查看协议栈，它采用分层方式描述一个完整的协议或一组协议（见图 1）。

帧/数据包封装

要理解以太网的工作原理，首先需要了解数据包封装的概念，以及协议栈如何融入这一概念。

协议栈的每一层都负责特定级别的功能。例如，物理层负责比特在介质上的实际电气传输。模型中的每一较高层都以某种独立的方式利用底层层（意味着层与层之间功能几乎没有重叠）。

这种分层方法通过封装来实现。这一概念最好通过图 2 所示的示例来说明。该示例展示了与 Web 浏览器会话相关的每一层如何映射到协议栈模型。

从应用层开始，Web 浏览器会使用特定于应用的命令生成一个 HTTP 请求。该请求随后被传递给 TCP 层，TCP 层会构建一个 TCP 数据包，该数据包由 TCP 头部和 TCP 数据组成。TCP 头部包含特定于 TCP 协议的信息，例如数据包序列信息、校验和信息以及源端口号和目的端口号（HTTP 通常使用 80 端口）。

在 IP 协议层，构造一个 IP 数据报以承载 TCP 数据包。与 TCP 数据包类似，IP 数据报由 IP 头部和 IP 数据组成。IP 头部包含服务类型、校验和、协议类型（TCP 为 06h）以及源和目的 IP 地址等信息。IP 数据报的数据域包含待传输的完整 TCP 数据包。

在数据链路层/物理层，IP 数据报使用 IEEE 802.3 协议在网络中传输。MAC（IEEE 802.3）帧由 MAC 头部和 MAC 载荷（数据）组成。MAC 头部包含有关 MAC 帧的信息，例如源 MAC 地址、目的 MAC 地址和帧长度。载荷字段包含待传输的完整 IP 数据报。

注意，每个协议中封装的各种地址各不相同，通常彼此之间没有固定的关系。在我们的示例中，TCP 数据包使用端口号，该端口号通常根据应用层协议分配（例如，HTTP 使用 80 端口）。IP 数据报使用 IP 地址，该地址通常从可用的互联网地址池中静态或动态分配，而 MAC 帧使用 MAC 地址，这些地址分配给特定的硬件设备。

注意 1: 本文档中，“MAC 帧”，“以太网帧”和 “IEEE 802.3 帧” 可互换使用。

2: 术语 “包”，“帧” 和 “数据报” 经常互换使用。这些术语适用于特定协议，例如 IEEE 802.3 帧、TCP 包或 IP 数据报。

图 1：互联网协议栈

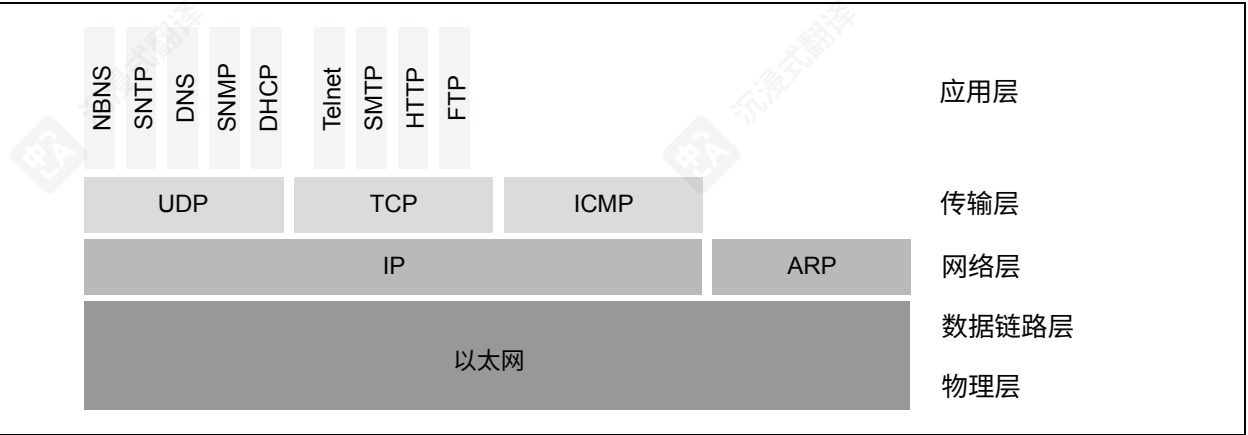
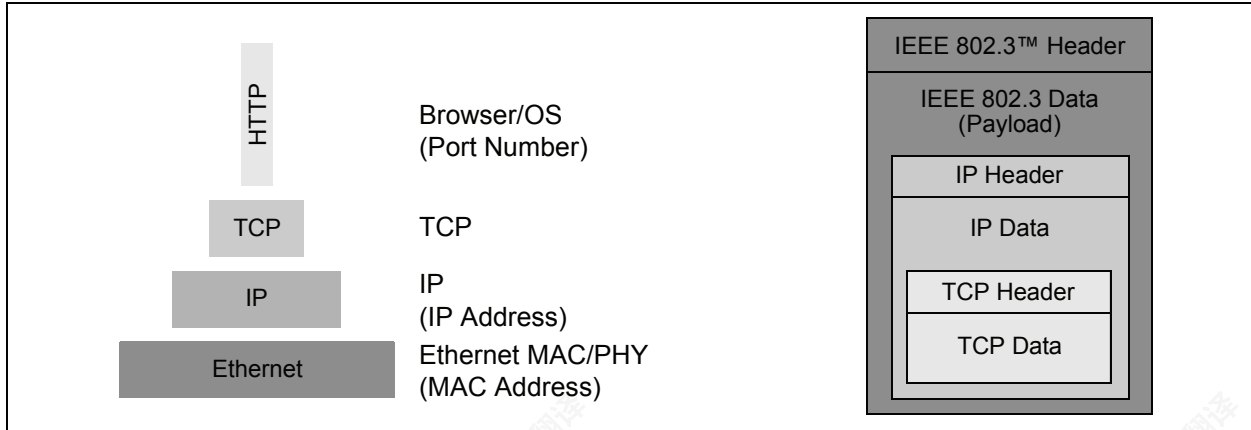


FIGURE 2: DATA ENCAPSULATION EXAMPLE



Application Layer Protocols

The application layer provides the user interface.

When used on top of some lower layer protocols (UDP or TCP – “**Transport Layer Protocols**” section), application layer protocols are usually assigned a port number. For example, HTTP servers are typically associated with port 80.

The following are common application layer protocols associated with the Internet:

Hyper Text Transfer Protocol (HTTP): Used primarily to transfer data associated with browsing of the World Wide Web.

Simple Mail Transfer Protocol (SMTP): Used to transport e-mails across the internet.

File Transfer Protocol (FTP): Used to transfer files or other pieces of data over the internet.

Domain Name System (DNS): Used to translate domain names, such as “microchip.com” into IP addresses.

Dynamic Host Configuration Protocol (DHCP): Used to dynamically assign IP addresses to a particular node from a pool of available IP addresses.

Telnet: Used to establish an interactive TCP connection to a node.

Simple Network Time Protocol (SNTP): Used to allow nodes to synchronize their clocks to a reference clock.

Simple Network Management Protocol (SNMP): Used to monitor network attached devices for conditions that require intervention, such as Faults, etc.

Transport Layer Protocols

The transport layer hides network dependent details from the layers above, including transport address to network address translation, sequencing, error detection/recovery, etc.

When used on top of the IP protocol, transport layer protocols are typically assigned an IP protocol number.

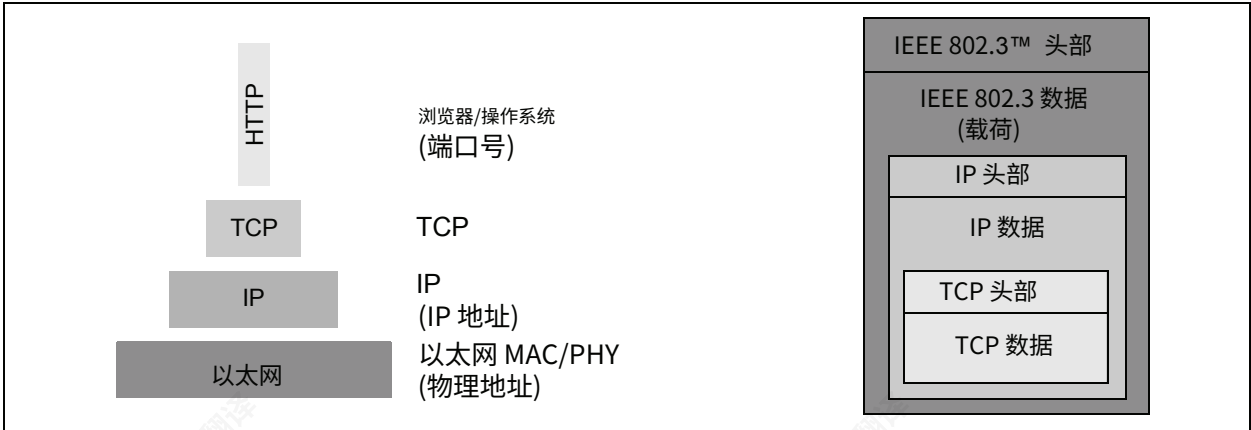
The following are common transport layer protocols associated with the internet:

Transmission Control Protocol (TCP): Provides reliable communication to applications.

User Datagram Protocol (UDP): Provides high performance, but unreliable communication to applications.

Internet Control Message Protocol (ICMP): Used to send network and/or node error or status messages.

图 2：数据封装示例



应用层协议

应用层提供用户界面。

当用于某些底层协议（UDP 或 TCP – “**Transport Layer Protocols**” 部分），应用层协议通常被分配一个端口号。例如，HTTP 服务器通常

与 80 端口关联。

以下是与互联网相关的常见应用层协议：

超文本传输协议（HTTP）：主要用于传输与万维网浏览相关的数据。

简单邮件传输协议（SMTP）：用于在互联网上传输电子邮件。

文件传输协议（FTP）：用于在互联网上传输文件或数据。

域名系统（DNS）：用于将域名（如 “microchip.com”）转换为 IP 地址。

动态主机配置协议（DHCP）：用于从可用 IP 地址池中动态向特定节点分配 IP 地址。Telnet：用于与节点建立交互式 TCP 连接。

简单网络时间协议（SNTP）：用于允许节点将其时钟与参考时钟同步。

简单网络管理协议（SNMP）：用于监控联网设备的状态，以便在需要干预时（例如故障）采取行动。

传输层协议

传输层向上层隐藏了网络相关的细节，包括传输地址到网络地址的转换、序列号、错误检测/恢复等。

当用于 IP 协议之上时，传输层协议通常被分配一个 IP 协议号。

以下是与互联网常见的传输层协议：

传输控制协议（TCP）：为应用程序提供可靠的通信。

用户数据报协议（UDP）：提供高性能，
but
不可靠的通信至应用程序。

互联网控制报文协议（ICMP）：用于发送网络和/或节点错误或状态消息。

Network Layer Protocols

The network layer determines how messages are routed in a network, including QoS (Quality of Service) services, provision of network addresses for the transport layer, etc.

When used on top of Ethernet, network layer protocols are typically assigned an “EtherType”, which is discussed in more detail in the “Ethernet Frame Format” section.

The following are common network layer protocols associated with the internet:

Address Resolution Protocol (ARP): Used to translate protocol addresses to hardware interface addresses, such as an IP address to a MAC address.

Reverse Address Resolution Protocol (RARP): Used to translate hardware interface addresses to protocol addresses, such as a MAC address to an IP address.

Internet Protocol (IP): Connectionless network layer protocol used by TCP, UDP, etc.

Physical/Data Link Layer Protocols

The physical layer provides for the transparent transmission of bit streams across physical connections, including encoding, multiplexing, synchronization, clock recovery, serialization, etc.

The data link layer is concerned with the transmission of frames (blocks) in an error-free manner, including frame sequencing, frame flow control, etc.

Ethernet is one of the most common physical/data link layer protocols, and the subject of the remainder of this application note.

PHYSICAL MEDIUM OVERVIEW

As mentioned previously, Ethernet is defined in part by the physical medium over which frames are transmitted. The following is a summary of the more common mediums:

- 1 Mb/s
 - 1Base5: 2 twisted telephone wire pairs
- 10 Mb/s
 - 10Broad36: 1 broadband cable
 - 10Base2: RG 58 coax cable
 - 10Base5: 1 coax cable
 - 10Base-F: 1 optical fiber
 - 10Base-T: 2 pairs UTP CAT3 or better, full-duplex
- 100 Mb/s
 - 100Base-FX: 2 optical fibers, Full-Duplex
 - 100Base-T2: 2 pairs UTP CAT3 or better, full-duplex
 - 100Base-T4: 4 pairs UTP CAT3 or better, half-duplex
 - 100Base-TX: 2 pairs UTP CAT5 or better, full-duplex
- 1 Gb/s
 - 1000Base-CX: Copper jumper cable
 - 1000Base-LX: Long wavelength Multi/Single mode fiber
 - 1000Base-SX: Short wavelength Multi mode fiber
 - 1000Base-T: 4 CAT5e, CAT6 or better pairs

Note 1: UTP – Unshielded Twisted Pair wire
2: CAT3 wires and copper telephone wires are essentially interchangeable.

网络层协议

网络层决定消息在网络中的路由方式，包括服务质量（QoS）服务、为传输层提供网络地址等。

当在以太网之上使用时，网络层协议通常被分配一个 “EtherType”，该 “以太网帧格式”部分将更详细地讨论它。

以下是与互联网相关的常见网络层协议：

地址解析协议（ARP）：用于将协议地址转换为硬件接口地址，例如将 IP 地址转换为 MAC 地址。

反向地址解析协议 (RARP)：用于将硬件接口地址转换为协议地址，例如将 MAC 地址转换为 IP 地址。

互联网协议 (IP)：TCP、UDP 等使用的无连接网络层协议。

物理/数据链路层协议

物理层提供在物理连接上透明传输比特流的功能，包括编码、复用、同步、时钟恢复、串行化等。

数据链路层负责以无差错的方式传输帧（块），包括帧序号、帧流控等。

以太网是最常见的物理层/数据链路层协议之一，也是本应用笔记的其余部分的主题。

物理介质概览

如前所述，以太网的部分定义在于帧传输的物理介质。以下是常见介质的总结：

- 1 MB/s
 - 1Base5：2 对双绞电话线
- 10 MB/s
 - 10Broad36: 1 条宽带电缆
 - 10Base2：RG 58 同轴电缆
 - 10Base5：1 根同轴电缆
 - 10Base-F：1 根光纤
 - 10Base-T: 2 对 UTP CAT3 或更高，全双工
- 100 MB/s
 - 100Base-FX: 2 根光纤，全双工
 - 100Base-T2: 2 对 UTP CAT3 或更高，全双工
 - 100Base-T4: 4 对 UTP CAT3 或更高，半双工
 - 100Base-TX: 2 对 UTP CAT5 或更高，全双工
- 1 Gbps
 - 1000Base-CX: 铜跳线电缆
 - 1000Base-LX: 长波长多模/单模光纤
 - 1000Base-SX: 短波长多模光纤
 - 1000Base-T: 4 对 CAT5e、CAT6 或更优线缆

注意 1: UTP – 非屏蔽双绞线
2: CAT3 线和铜电话线本质上可以互换使用。

ETHERNET SPECIFICATIONS

The Ethernet specification (IEEE 802.3) has evolved over the last number of years to address higher transmission rates and new functionality. Table 4 shows the most common specification supplements.

ETHERNET FRAME FORMAT

A basic 10/100 Ethernet frame consists of the following fields, as shown in Figure 3.

Preamble: Seven octets of 55h. In 100 Mb/s operation, the first octet is 4B/5B encoded to /J/K/ (more on what this means later), and is known as the Start-of-Stream Delimiter (SSD). The preamble is present to allow the receiver to lock onto the stream of data before the actual frame arrives.

Start-of-Frame Delimiter (SFD): ‘10101011b’ (as seen on the physical medium). The SFD is sometimes considered to be part of the preamble. This is why the preamble is sometimes described as eight octets.

Destination Address (DA): The 6-octet MAC address of the destination hardware. Please refer to the “**MAC Addresses**” section for information on multicast and broadcast addressing.

Source Address (SA): The 6-octet MAC address of the source hardware.

Length/Type: If the value in this 2-octet field is ≤ 1500 (decimal), this represents the number of octets in the payload. If the value is ≥ 1536, this represents the EtherType (payload type). The following are the most common EtherType values:

- IPv4 = 0800h
- IPv6 = 86DDh
- ARP = 0806h
- RARP = 8035h

Payload (Client Data): The client data, such as an IP datagram, etc. The minimum payload size is 46 octets; the maximum payload size is 1500 octets. While payloads below or above these limits do not meet the IEEE 802.3 specification, there is varied support for these payloads depending on the particular vendor. Please refer to the “**Frame Size**” section for further discussion on this topic.

Pad: Since the minimum payload size is 46 octets, pad octets must be inserted to reach this minimum if the payload size is less than 46 octets.

Frame Check Sequence (FCS): The value of the 4-octet FCS field is calculated over the source address, destination address, length/type, data and pad fields using a 32-bit Cyclic Redundancy Check (CRC).

End-of-Stream Delimiter (ESD): In 100 Mb/s operation, the PHY transmits a /T/R/ symbol pair after the FCS (during the inter-frame gap) to denote the end of the frame.

In 10 Mb/s operation, a special TP_IDL signal (discussed later in this document) and network silence indicates the end of the frame. Like the /T/R/ symbol pair in 100Base-T, this special TP_IDL marker is not considered part of the frame data.

Note: MAC frames are enumerated in terms of “octets” (one octet = 8 bits).

FIGURE 3: BASIC FRAME FORMAT

10/100 IEEE 802.3™ Frame	
7 octets	Preamble
1 octet	Start Frame Delimiter (SFD)
6 octets	Destination Address (DA)
6 octets	Source Address (SA)
2 octets	Length (≤ 1500) Type (≥ 1536)
46 octets to 1500 octets	Client Data (Payload)
4 octets	Pad (if necessary)
	Frame Check Sequence (FCS)

Besides the basic frame described above, there are two other common frame types in 10/100 Ethernet: control frames and VLAN tagged frames. Figure 4 shows a comparison between the three common 10/100 frame formats and the gigabit Ethernet frame format.

以太网规范

以太网规范（IEEE 802.3）在过去几年中不断发展，以应对更高的传输速率和新功能。表 4 显示了最常见的规范补充。

以太网帧格式

一个基本的 10/100 以太网帧包含以下字段，如图 3 所示。

前言：7 个字节为 55h。在 100 Mb/s 操作模式下，第一个字节经过 4B/5B 编码为 /J/K/（稍后详述其含义），该字节被称为流起始分隔符（SSD）。前言的存在是为了让接收方在数据帧到达之前能够锁定数据流。

帧起始符 (SFD): ‘10101011b’（在物理介质上所见）。SFD 有时被视为前导码的一部分。这就是为什么前导码有时被描述为八个字节。

目的地址 (DA)：目的硬件的 6 字节 MAC 地址。有关组播和广播地址的信息，请参阅“MAC 地址”部分。

源地址 (SA)：源硬件的 6 字节 MAC 地址。

长度/类型：如果此 2 字节字段中的值为 ≤ 1500（十进制），则表示载荷中的字节数。如果值为 ≥ 1536，则表示 EtherType（载荷类型）。以下是最常见的 EtherType 值：

- IPv4 = 0800h
- IPv6 = 86DDh
- ARP = 0806h
- RARP = 8035h

载荷 (客户端数据 (载荷)): 客户端数据，例如 IP 数据报等。最小载荷大小为 46 字节；最大载荷大小为 1500 字节。虽然低于或高于这些限制的载荷不符合 IEEE 802.3 规范，但不同厂商对这些载荷的支持情况各不相同。有关此主题的进一步讨论，请参阅“Frame Size”部分。

Pad: 由于最小载荷大小为 46 字节，如果载荷大小小于 46 字节，则必须插入填充字节以达到该最小值。

帧校验序列 (FCS)：4 字节 FCS 字段的值是使用 32 位循环冗余校验 (CRC) 针对源地址、目的地址、长度/类型、数据及填充字段进行计算的。

流结束符 (ESD): 在 100 Mb/s 操作模式下，物理层在帧校验序列 (FCS) 之后（在帧间隙期间）发送一个 /T/R/符号对，以表示帧的结束。

在 10 Mb/s 操作模式下，特殊的 TP_IDL 信号（本文稍后讨论）和网络静默表示帧的结束。与 100Base-T 中的 /T/R/符号对类似，这个特殊的 TP_IDL 标记不被视为帧数据的一部分。

注意：MAC 帧以“字节”（一个字节 = 8 位）为单位进行枚举。

图 3：基本框架格式

10/100 IEEE 802.3™ 帧	
7 字节	序言
1 字节	起始帧分隔符 (SFD)
6 字节	目的地址 (DA)
6 字节	源地址 (SA)
2 字节	长度 (≤ 1500) 类型 (≥ 1536)
46 字节 to 1500 字节	客户端数据 (载荷)
4 字节	垫 (如有必要)
	帧校验序列 (FCS)

除了上述基本帧之外，10/100 以太网还有两种常见的帧类型：控制帧和 VLAN 标记帧。图 4 展示了三种常见的 10/100 帧格式与千兆以太网帧格式之间的对比。

FIGURE 4: COMMON ETHERNET FRAME TYPES

	10/100 Data Frame	10/100 Control Frame	10/100 VLAN Frame	Gigabit Data Frame
7 octets	Preamble	Preamble	Preamble	Preamble
1 octet	Start Frame Delimiter (SFD)	Start Frame Delimiter (SFD)	Start Frame Delimiter (SFD)	Start Frame Delimiter (SFD)
6 octets	Destination Address (DA)	Destination Address (DA)	Destination Address (DA)	Destination Address (DA)
6 octets	Source Address (SA)	Source Address (SA)	Source Address (SA)	Source Address (SA)
2 octets	Length (≤ 1500) Type (≥ 1536)	8808h	8100h	Length (≤ 1500) Type (≥ 1536)
2 octets			Tag Control Information	
2 octets			Length (≤ 1500) Type (≥ 1536)	
46 octets to 1500 octets	Client Data (Payload)	Control Opcodes (2 octets) Control Parameters (2 octets) 00h (42 octets)	Client Data (Payload)	Client Data (Payload)
4 octets	Pad (if necessary)		Pad (if necessary)	Pad (if necessary)
0 octets to 448 octets	Frame Check Sequence (FCS)	Frame Check Sequence (FCS)	Frame Check Sequence (FCS)	Frame Check Sequence (FCS)
				Carrier Extension

FIGURE 4: COMMON ETHERNET FRAME TYPES

	10/100 Data Frame	10/100 Control Frame	10/100 VLAN Frame	Gigabit Data Frame
7 octets	Preamble	Preamble	Preamble	Preamble
1 octet	Start Frame Delimiter (SFD)	Start Frame Delimiter (SFD)	Start Frame Delimiter (SFD)	Start Frame Delimiter (SFD)
6 octets	Destination Address (DA)	Destination Address (DA)	Destination Address (DA)	Destination Address (DA)
6 octets	Source Address (SA)	Source Address (SA)	Source Address (SA)	Source Address (SA)
2 octets	Length (≤ 1500) Type (≥ 1536)	8808h	8100h	Length (≤ 1500) Type (≥ 1536)
2 octets			Tag Control Information	
2 octets			Length (≤ 1500) Type (≥ 1536)	
46 octets to 1500 octets	Client Data (Payload)	Control Opcodes (2 octets) Control Parameters (2 octets) 00h (42 octets)	Client Data (Payload)	Client Data (Payload)
4 octets	Pad (if necessary)		Pad (if necessary)	Pad (if necessary)
0 octets to 448 octets	Frame Check Sequence (FCS)	Frame Check Sequence (FCS)	Frame Check Sequence (FCS)	Frame Check Sequence (FCS)
				Carrier Extension

FRAME SIZE

When discussing IEEE 802.3 frame sizes, the Preamble/SFD is typically not included in the size of the frame. Therefore, the minimum and maximum allowed size of a basic or control frame is 64 octets and 1518 octets, respectively. Conversely, the maximum size for a VLAN tagged frame (described in the “**VLAN Tagged Frames**” section) is defined as 1522 octets.

Frames below the 64-octet limit are often known as “runt” frames, while frames above the 1518-octet limit are often known as “long” or “huge” frames. The term, “jumbo” frames, refers to frames above 1518 octets in 10/100Base-T and to 9000 octet frames in gigabit Ethernet. The term “giant” is sometimes used to refer to frames that are more than 6000 octets long.

In some literature, the term “frame size” refers solely to the payload of the frame. It is, therefore, common to see the term “jumbo frame” defined as a frame with a size of greater than 1500 octets.

Control Frames

Ethernet frames with an EtherType value of 8808h are specified as MAC control frames, and are used to control the flow of frames on a link. Implementation of MAC control features in an Ethernet node is optional.

The first two octets in a MAC control frame payload contain the opcode. Currently, the only standard control frame is a pause frame, which has an opcode and a destination address as follows:

- Opcode: 0001h
- Address: 01-80-c2-00-00-01 (multicast)

A pause frame requests that the station at the other end of the link stop transmitting for a period of time (specified by a 2-octet pause time after the opcode). One pause “quanta” is equal to 512 bit times.

Transmitting a pause frame with a pause time value of 0000h means to cancel any existing pauses in effect.

VLAN Tagged Frames

Virtual Local Area Network (VLAN) tagging adds additional information, known as tag control information, into the frame for the purpose of allowing the creation of networks defined by a logical topology, rather than a physical topology.

MAC ADDRESSES

A MAC address is a 48-bit (6-octet) number unique to every piece of Ethernet hardware. It consists of a 24-bit Organizationally Unique Identifier (OUI) and a 24-bit hardware identifier, as shown in Figure 5.

OUIs are assigned by the IEEE to a particular company or organization (Microchip’s OUI is 00-04-A3h), while hardware IDs are assigned by the owner of that particular OUI.

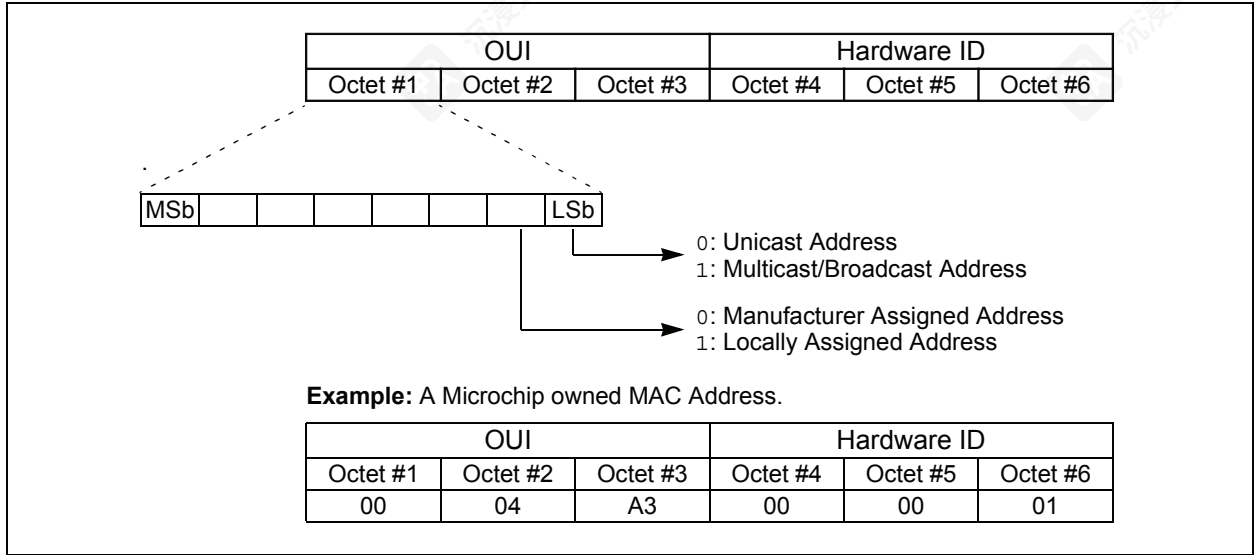
For individuals requiring 4096 MAC addresses or less, an Individual Address Block (IAB) may be purchased. An IAB consists of a reserved OUI (owned by the IEEE) plus 12 bits of reserved hardware identifier, which leaves 12 bits of hardware identifier available to the purchaser, for a total of 4096 unique MAC addresses.

MAC address octets are transmitted high-order (Octet #1) first, while bits within an octet are transmitted low-order, Least Significant bit (LSb) first.

A MAC address whose Least Significant bit of Octet #1 is set as a multicast address is intended for one or more nodes. As an example, pause frames, which have an address of 01-80-c2-00-00-01, are considered multicast packets.

A MAC address of FF-FF-FF-FF-FF-FF is a broadcast address, which is intended for all nodes.

FIGURE 5: MAC ADDRESSES



框架尺寸

在讨论 IEEE 802.3 帧大小时，前导/帧起始定界符通常不计入帧的大小。因此，基本帧或控制帧的最小和最大允许大小分别为 64 字节和 1518 字节。相反，VLAN 标记帧（见“VLAN Tagged Frames”部分）的最大大小定义为 1522 字节。

64 字节以下的帧通常被称为“runt”帧，而 1518 字节以上的帧通常被称为“long”或“huge”帧。术语“jumbo”帧指的是 10/100Base-T 中超过 1518 字节的帧以及千兆以太网中的 9000 字节帧。术语“giant”有时用于指代长度超过 6000 字节的帧。

在某些文献中，术语“frame size”仅指帧的负载部分。因此，常见将术语“jumbo frame”定义为大小超过 1500 字节“frame size”的帧。

控制帧

以太网帧的 EtherType 值为 8808h 的帧被指定为 MAC 控制帧，用于控制链路上的帧流。以太网节点中 MAC 控制功能的实现是可选的。

MAC 控制帧载荷中的前两个字节包含操作码。目前，唯一的标准控制帧是暂停帧，其操作码和目的地址如下：

- Opcode: 0001h
- 地址：01-80-c2-00-00-01（组播）

一个暂停帧请求链路另一端的站点停止传输一段时间（由 opcode 之后的 2 个字节指定的暂停时间）。一个暂停“量”等于 512 比特时间。

发送暂停帧，暂停时间值为 0000h 表示取消任何正在生效的暂停。

VLAN 标记帧

虚拟局域网（VLAN）打标签会将称为标签控制信息的附加信息添加到帧中，以便创建由逻辑拓扑而非物理拓扑定义的网

MAC 地址

MAC 地址是 48 位（6 字节）数字，专用于每一块以太网硬件。它由 24 位组织唯一标识符（OUI）和 24 位硬件标识符组成，如图 5 所示。

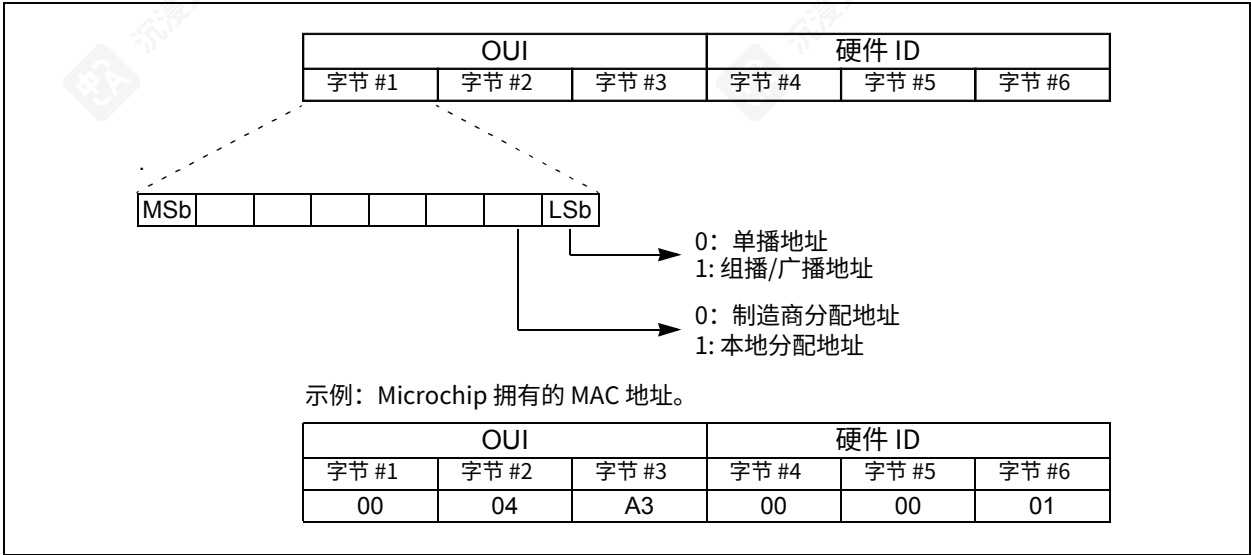
OUI 由 IEEE 分配给特定公司或组织（Microchip’的 OUI 为 00-04-A3h），而硬件 ID 由该特定 OUI 的所有者分配。

对于需要 4096 个或更少 MAC 地址的个人用户，可以购买一个个人地址块（IAB）。IAB 由一个保留的 OUI（由 IEEE 拥有）加上 12 位保留的硬件标识符组成，这使得购买者拥有 12 位硬件标识符可用，总共可生成 4096 个唯一的 MAC 地址。

MAC 地址的字节按高位（第 1 个字节）先传输，而字节内的位按低位、最低有效位（LSb）先传输。

第 1 个字节的少数有效位设置为组播地址的 MAC 地址旨在用于一个或多个节点。作为考试具有地址为 01-80-c2-00-00-01, are 考虑组播数据包。01-80-c2-00-00-01, are 考虑 FF-FF-FF-FF-FF-FF 的 MAC 地址是广播地址，旨在发送给所有节点。

图 5：MAC 地址



STREAM CONSTRUCTION/
DECONSTRUCTION

Based on the previous discussion of the protocol layer model and frame encapsulation, we are now ready to discuss the functions of the Ethernet MAC and PHY. The IEEE 802.3 definition of the PHY and MAC layers for 100 Mb/s are shown in Figure 6. What is important to realize from this diagram is that the functions of the Ethernet PHY and MAC, and the interfaces of each, are defined by the IEEE 802.3 specification.

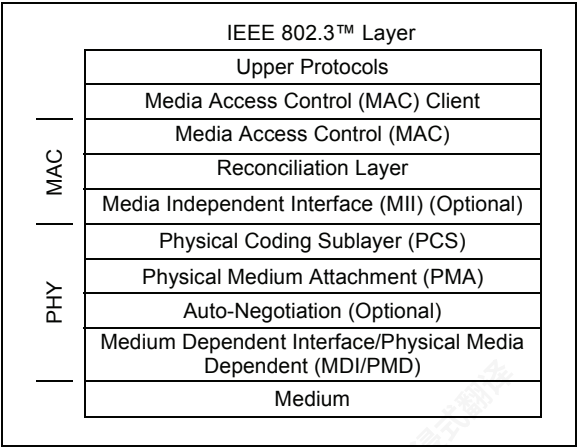
The physical interface to the transmission medium is called the MDI, and changes depending on which medium (twisted pair, fiber, etc.) is used.

The interface between the PHY and the MAC is called the MII, and is composed of a receive path, a transmit path and a management path, which is used to read and write PHY registers. The width of the receive and transmit paths are the same, and is determined by the speed that the MAC and PHY are implementing, as follows:

- 10 Mb/s: 4 bits wide at 2.5 MHz
- 100 Mb/s: 4 bits wide at 25 MHz

Note: There are also Reduced MII (RMII) and Serial MII (SMII) interfaces defined that are 2-bit and 1-bit wide, respectively.

FIGURE 6: IEEE 802.3™ 100 Mb/s
LAYER DEFINITIONS



Reconciliation Layer: Maps the physical status (carrier loss, collision, etc.) to the MAC layer.

Media Independent Interface (MII) (Optional): Provides an n-bit transmit/receive interface to the PHY.

Physical Coding Sublayer (PCS): Encoding, multiplexing and synchronization of outgoing symbol streams (4B/5B encoding, etc.).

Physical Medium Attachment (PMA): Signal transmitter/receiver (serialization/deserialization of symbol stream, clock recovery, etc.).

Auto-Negotiation (Optional): Negotiation to the highest mode supported by both hosts.

Medium Dependent Interface/Physical Media Dependent (MDI/PMD): RJ45, etc.

Medium: UTP, Fiber, etc.

流式构建/解构

基于之前对协议层模型和帧封装的讨论，我们现在可以讨论以太网 MAC 和 PHY 的功能。IEEE 802.3 对 100 Mb/s 的 PHY 和 MAC 层的定义如图 6 所示。从该图中重要的是要认识到，以太网 PHY 和 MAC 的功能以及各自的接口是由 IEEE 802.3 标准定义的。

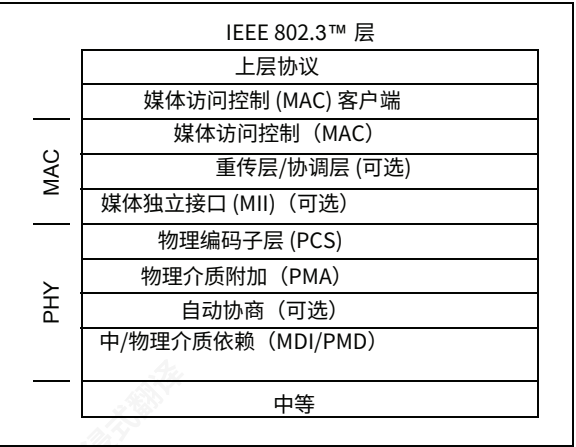
传输介质的物理接口称为 MDI，其具体形式取决于所使用的介质（如双绞线、光纤等）。

物理层与 MAC 之间的接口称为 MII，由接收路径、发送路径和管理路径组成，该管理路径用于读取和写入 PHY 寄存器。接收路径和发送路径的宽度相同，由 MAC 和 PHY 所实现的速度决定，具体如下：

- 10 Mb/s: 2.5 MHz 下 4 位宽
- 100 Mb/s: 25 MHz 下 4 位宽

注意：还定义了两种接口，分别是 2 位宽的 Reduced MII (RMII) 和 1 位宽的 Serial MII (SMII)。

图 6: IEEE 802.3™ 100 Mb/s
层定义



重传层/协调层：映射物理状态 (载波损耗，碰撞等) 到 MAC 层。
媒体独立接口 (MII) (可选)：提供 n 位发送/接收接口到 PHY。

Physical Coding Sublayer (PCS): 编码 g, 输出符号流的复用与同步 (4B/5B 编码等)。

物理 中等 附件 (PMA): 信号 of 发射器/接收器 (序列化/反序列化) 符号流，时钟恢复等)。

自动协商 (可选)：协商到双方主机都支持的最高模式。

中等 依赖 接口/物理 媒体 从属 (MDI/PMD)：RJ45 等。
介质：UTP、光纤等。

FIGURE 7: STREAM DECONSTRUCTION (RX)

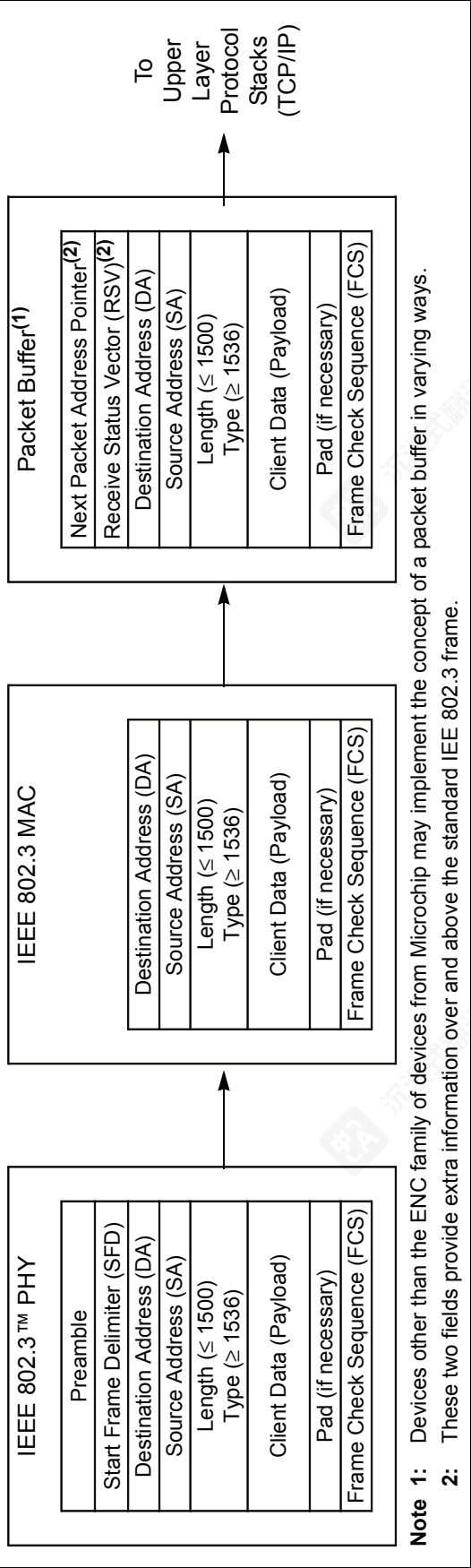


FIGURE 8: STREAM CONSTRUCTION (TX)

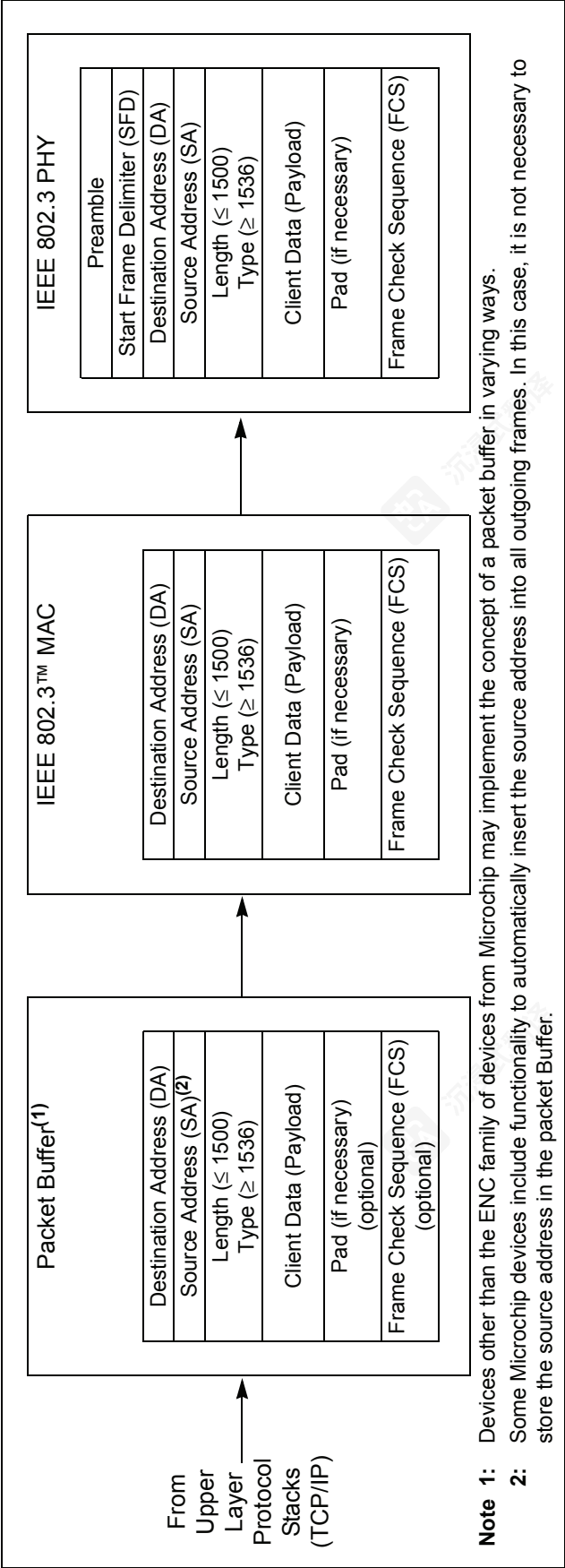


FIGURE 7: STREAM DECONSTRUCTION (RX)

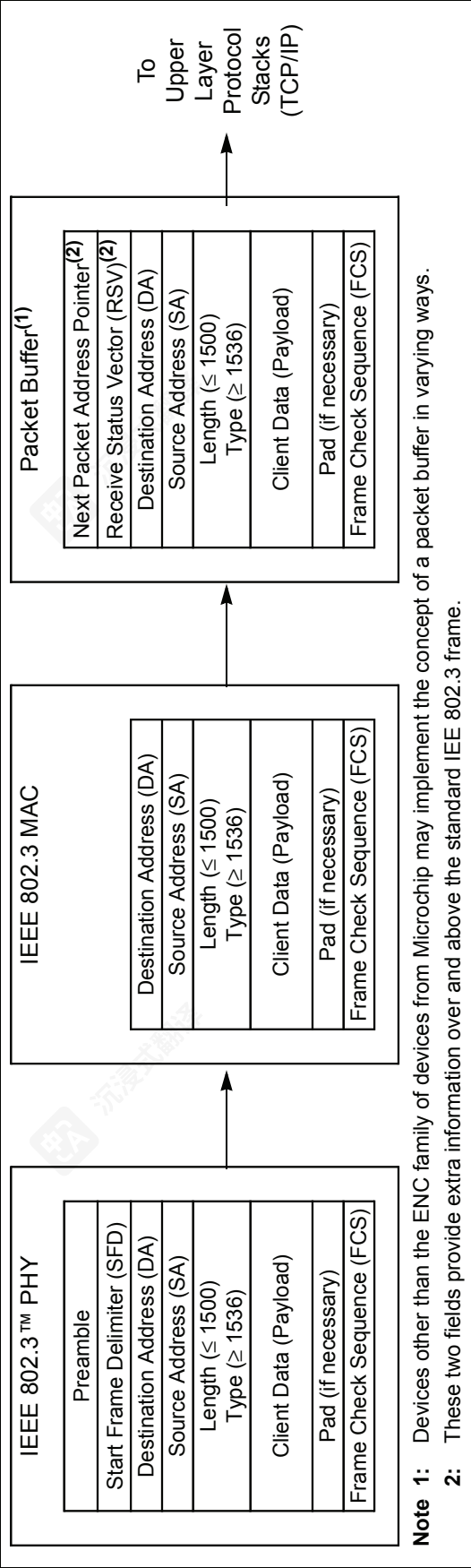
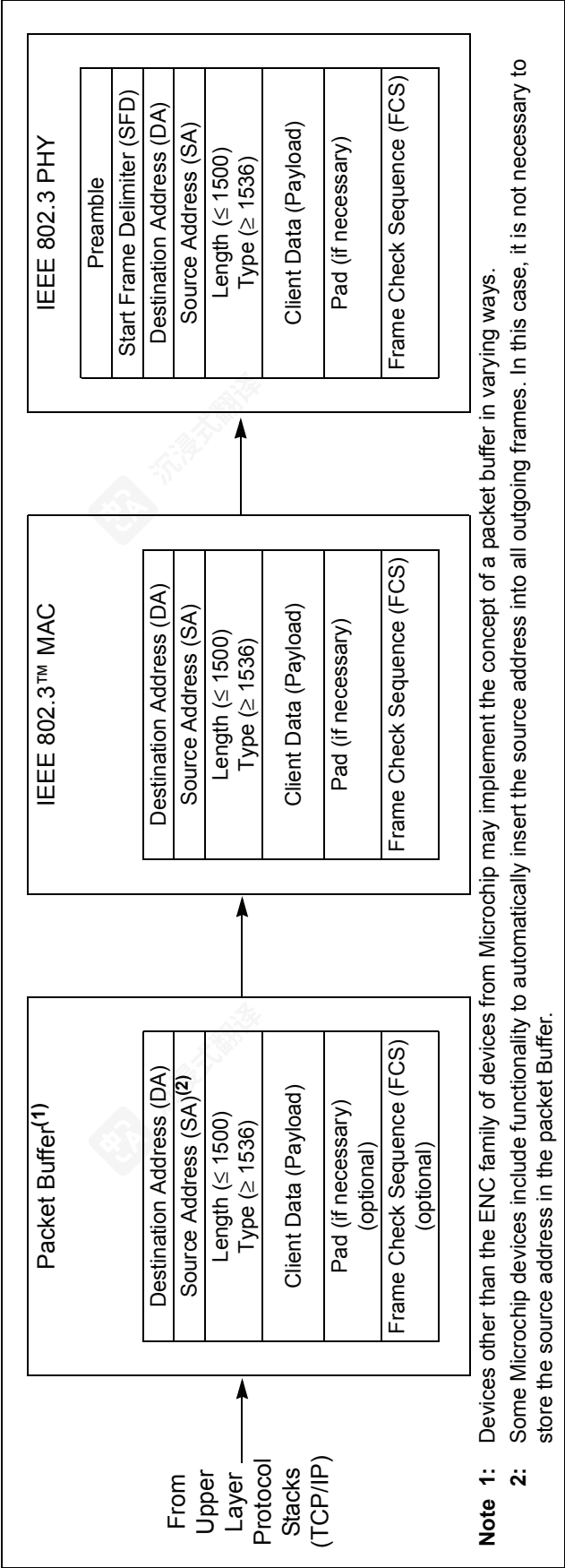


FIGURE 8: STREAM CONSTRUCTION (TX)



STREAM TIMING

So far, we have discussed how data is assembled and disassembled into packets, and the role of the MAC and the PHY in doing so. What still remains is the actual transmission of the constructed stream over the physical medium.

Before we can understand the timing of IEEE 802.3 frames, we have to understand the reasons behind the timing.

Carrier Sense Multiple Access with Collision Detect (CSMA/CD)

Originally, Ethernet was designed as a protocol to run over a shared medium, as shown in Figure 9. In this topology, each node on the bus has equal access to the bus, but only one node may transmit at a time, and each node transmits half-duplex. Simultaneous transmission from multiple nodes would result in garbled data on the medium, and subsequent loss of data. From this simple example, we can derive some basic requirements for a network protocol:

- Multiple nodes must be able to transmit on a shared medium (Multiple Access).
- Each node must be able to detect when another node is transmitting (Carrier Sense).
- A transmitting node must be able to determine when simultaneous transmission occurs in the case where multiple nodes see the medium as Idle and start transmitting at the same time (Collision Detect).
- When a collision is detected, each node must have a method to determine when to retransmit without each node continually trying to retransmit at the same time (Backoff).

These requirements are met in Ethernet using a scheme known as Carrier Sense Multiple Access with Collision Detect (CSMA/CD).

Before an Ethernet node can begin transmitting, it must first determine whether the medium is active or Idle (Carrier Sense). If the medium is active, then that node must wait until the medium becomes Idle, and then waits a predetermined amount of time after that before starting to transmit. This predetermined amount of time is called the Inter-Packet Gap (IPG), also known as an Inter-Frame Gap (IFG), and is dependent on the speed of the bus, as shown in Table 2. The IPG is used as a recovery time between frames to allow nodes to prepare for reception of the next frame.

However, if multiple nodes are waiting for the medium to become Idle, then they may start transmitting at virtually the same time once the medium becomes Idle. Therefore, all nodes must also have the ability to detect these collisions (Collision Detect).

If the nodes that are trying to transmit on an Idle medium are at physically opposite ends of the medium, and one node starts transmitting just before it sees the transmission from the other node on the medium, then the worst case scenario occurs. As an example, let us assume that Node 1 and Node 4 in Figure 9 both want to transmit. Node 1 starts to transmit, but the data takes some time to propagate down the medium to Node 4. Node 4 starts to transmit just before it sees the data from Node 1. Node 4 will almost immediately detect a collision on the medium, and will transmit a special pattern known as a jam signal onto the medium. This jam signal must now return to Node 1 before it can detect that a collision has occurred by comparing its transmitted data to the received data. This applies to 10Base2, 10Base5 and 10Base-F nodes, where all nodes share a common medium.

FIGURE 9: SHARED BUS TOPOLOGY (10Base2)

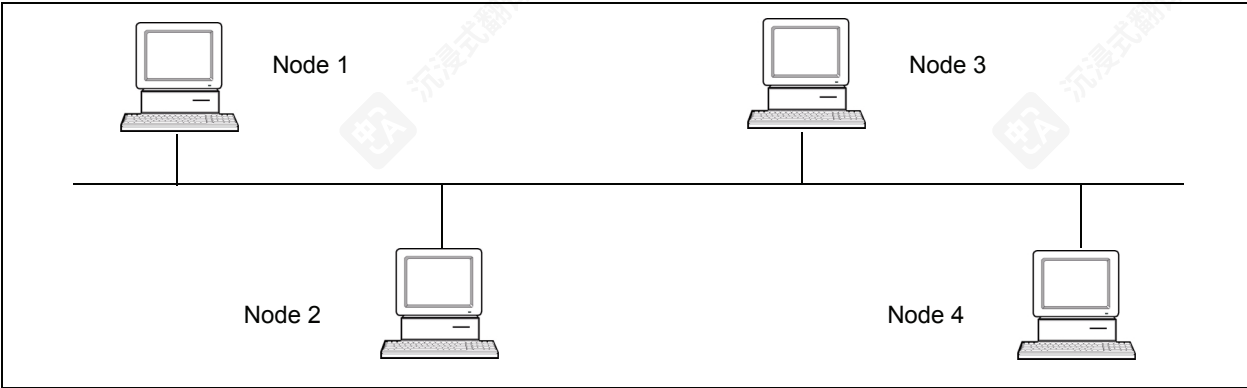


TABLE 2: KEY ETHERNET TIMING PARAMETERS

	Bit Time	IPG Time	Slot Time	Network Diameter (without repeaters)	Network Diameter (with repeaters)
10Base-T	100 ns	9.6 μs	512 Bit Times = 51.2 μs	100m	2500m
100Base-TX	10 ns	960 ns	512 Bit Times = 5.12 μs	100m	205m

流时序

到目前为止，我们讨论了数据如何被组装和拆分为数据包，以及 MAC 和 PHY 在此过程中的作用。尚待讨论的是构建好的流如何在物理介质上进行实际传输。

在理解 IEEE 802.3 帧的时序之前，我们必须了解时序背后的原因。

载波监听多路访问/冲突检测协议 (CSMA/CD)

最初，以太网被设计为一种在共享介质上运行的协议，如图 9 所示。在此拓扑结构中，总线上的每个节点对总线具有平等的访问权限，但同一时间只能有一个节点进行传输，且每个节点均为半双工模式。多个节点同时传输会导致介质上出现乱码数据，并随后造成数据丢失。从这个简单的例子中，我们可以推导出网络协议的一些基本要求：

- 多个节点必须能够在共享介质上发送数据（多路访问）。
- 每个节点必须能够检测到另一个节点正在发送信号（载波侦听）。
- 在多个节点将信道视为空闲并同时在同一时刻开始发送的情况下，发送节点必须能够确定是否发生了同时发送（碰撞检测）。
- 当检测到碰撞时，每个节点必须有一种方法来确定何时重传，以避免所有节点同时尝试重传（退避）。

这些要求通过一种称为载波监听多路访问/冲突检测协议（CSMA/CD）的方案在以太网中得到满足。

在以太网节点开始传输之前，它必须首先确定介质是否处于活动状态或空闲状态（载波监听）。如果介质处于活动状态，则该节点必须等待直到介质变为空闲状态，然后再等待一段预定的时间之后才开始传输。这段预定的时间称为帧间间隔（IPG），也称为帧间间隙（IFG），其值取决于总线速度，如表 2 所示。IPG 用作帧之间的恢复时间，以便节点为接收下一帧做好准备。

然而，如果有多个节点正在等待介质变为空闲状态，那么一旦介质变为空闲，它们可能会几乎同时开始传输。因此，所有节点也必须具备检测这些碰撞的能力（碰撞检测）。

如果试图在空闲介质上传输的节点位于介质的物理两端，且一个节点在尚未看到另一个节点在介质上的传输之前就开始传输，则会出现最坏情况。例如，假设图 9 中的节点 1 和节点 4 都想传输。节点 1 开始传输，但数据需要一定时间才能沿介质传播到节点 4。节点 4 在尚未看到数据之前就开始传输

从节点 1 开始。节点 4 将几乎立即检测到介质上的冲突，并将发送一个特殊的 pat- 被称为碰撞信号，然后才能通过将发送数据与接收数据进行比较来检测是否发生冲突。此规则适用于 10Base2，

10Base5 和 10Base-F 节点，其中所有节点共享一种通用的媒介。

图 9：共享总线拓扑（10Base2）

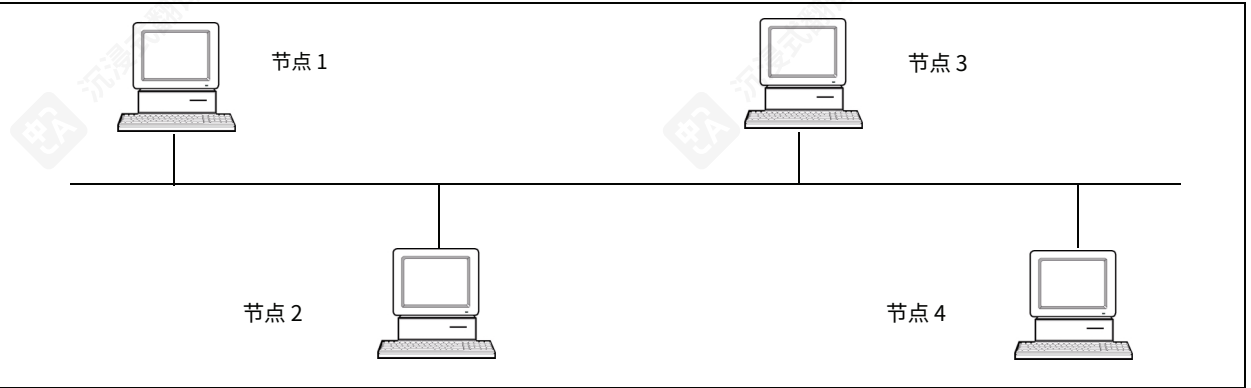


表 2：以太网关键时序参数

	位时间	IPG 时间	槽时间	网络直径 (含中继器)	网络直径 (无中继器)
10Base-T	100 纳秒	9.6 μ 个	512 位时间 = 51.2 μ 秒	100m	2500 米
100Base-TX	10 纳秒	960 纳秒	512 位时间 = 5.12 μ 秒	100m	205m

This means roughly two times the signal propagation time of the network can occur before all nodes on the medium are ensured to have detected it. This time is known as the collision window or slot time. The slot times for various speeds are shown in Table 2.

The situation we have described is known as an “in-window” collision because the collision is detected within the slot time. If, however, the size of the network is larger than the network diameter, an “out-of-window” or “late” collision can occur. Late collisions are not considered a transmission error like in-window collisions, but are instead considered a problem with the network topology itself. Unlike in-window collisions, late collisions are not dealt with at the physical/data link layers of Ethernet, but rather must be detected and handled by the application software.

Based on the above example, it should be somewhat evident that the collision window is equivalent to the minimum size of the frame. However, increasing the frame size then increases the impact of recovering from a collision.

To this end, the original authors of the IEEE 802.3 specification compromised by coming up with a “reasonable” collision window (referred to as the “Network Diameter” in Table 2) for 10Base-T and 100Base-T Ethernet. The minimum frame size was then set to match the chosen network diameter. It would follow naturally that gigabit Ethernet, which runs at 1000 Mb/s, would have a network diameter 1/10 that of 100Base-T. However, this would result in a practically unusable network diameter of about 20m. Gigabit Ethernet extends the frame size by adding bits at the end of the frame (called “Carrier Extension”) to form an effective minimum frame length of 4096 bits. This results in a network diameter roughly the same as for 100Base-T.

Since the transmission rate for 100Base-T is 10 times as fast as the transmission rate for 10Base-T, the time required to transmit a frame is 1/10 the time. This, in turn, means the slot time is reduced from about 50 μs for 10Base-T to about 5 μs for 100Base-T. Consequently, the network diameter shrinks from 2500m to about 200m.

Note that half-duplex can be used on topologies that do not use a shared bus topology, such as a point-to-point connection (Figure 10). In this case, the TX line of one node is connected to the RX line of the other node, and vice-versa. Consequently, a collision is much easier to detect, as each node can simply look for data on its RX port while it is transmitting. If any data is received while it is transmitting, the linked node must be transmitting as well, and a collision has occurred. This applies to 10Base-T and all 100 Mb/s and gigabit Ethernet nodes.

The last requirement for our network protocol is a method by which each node determines when to retransmit. If every node tries to retransmit at the same time, collisions would continue ad infinitum.

For this reason, Ethernet implements what is known as a binary exponential backoff algorithm, which works as follows:

1. Each node chooses a random delay (in the range from 0 to 1) before attempting its first retransmit.
2. If another collision occurs, each node doubles the range of random delays (now from 0 to 3) and chooses a random delay again.
3. This process repeats (with a range of 0 to 7, 0 to 15, etc.) until no collision occurs or until 10 attempts have been made. At this point, the defined range for each node will be 0 to 1023. In this manner, the range of backoff times increases exponentially with each try, and the probability of a collision rapidly decreases.
4. Six more attempts (for a total of 16 attempts) will be made to retransmit. If a node is still unsuccessful at retransmitting, the frame is dropped, and an excessive collision error is reported. The application software must then detect the dropping of the frame and try to retransmit the dropped frame, if desired.

这意味着在网络信号传播时间的两倍时间内，可以确保介质上的所有节点都已检测到该信号。这段时间被称为碰撞窗口或时隙时间。各种速率下的时隙时间见表 2。

我们描述的这种情况被称为“在窗口”内碰撞，因为碰撞是在时隙时间内被检测到的。然而，如果网络的大小大于网络直径，则可能发生“窗口外”或“晚”碰撞。晚碰撞不像在窗口内碰撞那样被视为传输错误，而是被视为网络拓扑本身的问题。与在窗口内碰撞不同，晚碰撞不在以太网的物理层/数据链路层处理，而必须由应用程序软件进行检测和处理。

基于上述示例，碰撞窗口应大致等同于帧的最小尺寸。然而，增大帧尺寸则会增加从碰撞中恢复的影响。

为此，IEEE 802.3 规范的原作者妥协，为 10Base-T 和 100Base-T 以太网制定了一个“合理”的冲突窗口（在表 2 中称为“网络直径”）。随后，最小帧大小被设定为与所选网络直径相匹配。由此自然可知，运行速率为 1000 Mb/s 的千兆以太网，其网络直径应为 100Base-T 的 1/10。然而，这将导致一个实际不可用的网络直径，约为 20 米。千兆以太网通过在帧末尾添加比特（称为“载波扩展”）来扩展帧大小，从而形成有效最小帧长度为 4096 比特。这使得网络直径大致与 100Base-T 相同。

由于 100Base-T 的传输速率是 10Base-T 传输速率的 10 倍，因此传输一个帧所需的时间是原来的 1/10。这，在切换，意味着时隙时间从 10Base-T 的约 50 μ 秒减少到 100Base-T 的约 5 μ 秒。因此，网络直径从 2500 米缩小到约 200 米。

注意，半双工模式也可用于不使用共享总线拓扑的拓扑结构中，例如点对点连接（图 10）。在这种情况下，一个节点的 TX 线连接到另一个节点的 RX 线，反之亦然。因此，冲突检测更容易，因为每个节点只需在发送时检查其 RX 端口是否有数据。如果在发送期间接收到任何数据，则说明相连的节点也在发送，此时发生了冲突。此规则适用于 10Base-T 以及所有 100 Mb/s 和千兆以太网节点。

我们网络协议的最后一个要求是：每个节点确定何时重传的方法。如果所有节点同时尝试重传，碰撞将无限持续下去。

因此，以太网实现了所谓的二进制指数退避算法，其工作原理如下：

1. 每个节点在尝试第一次重传之前，都会随机选择一个延迟（范围从 0 到 1）。
2. 如果发生另一次碰撞，每个节点将随机延迟的范围翻倍（现在从 0 到 3），并再次选择随机延迟。
3. 此过程重复执行（范围从 0 到 7，0 到 15，以此类推），直到不发生冲突或尝试次数达到 10 次为止。此时，每个节点定义的范围将为 0 到 1023。通过这种方式，每次重试的退避时间范围呈指数级增长，冲突发生的概率迅速降低。
4. 将额外进行六次重试（总计 16 次）。如果节点在重传时仍不成功，则丢弃该帧并报告过多的碰撞错误。应用程序软件必须检测帧的丢失，并在需要时尝试重传丢失的帧。

Full-Duplex Operation

While early Ethernet networks were implemented with a shared medium, and required CSMA/CD, most modern Ethernet networks are configured in a point-to-point (Figure 10) or a star topology (Figure 11), which can be thought of as a collection of point-to-point connections.

In either configuration, as each node is connected to a maximum of one other node, each node may operate in Full-Duplex mode. With a point-to-point/full-duplex configuration, collisions are not possible, and CSMA/CD is therefore not used. Each node may transmit whenever it wants to, within the constraints placed upon transmission by the inter-packet gap.

In addition, the total throughput of the medium is doubled (i.e., from 10 Mb/s to 20 Mb/s or from 100 Mb/s to 200 Mb/s).

Full-duplex operation also has the benefit of removing the limitations in network diameter due to slot times.

Note that not all medium types support full-duplex. In particular, the following types do not:

- 10Base2
- 10Base5
- 10Base-FP
- 10Base-FB
- 100Base-T4

FIGURE 10: POINT-TO-POINT TOPOLOGY

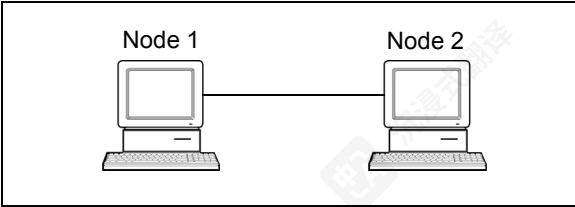
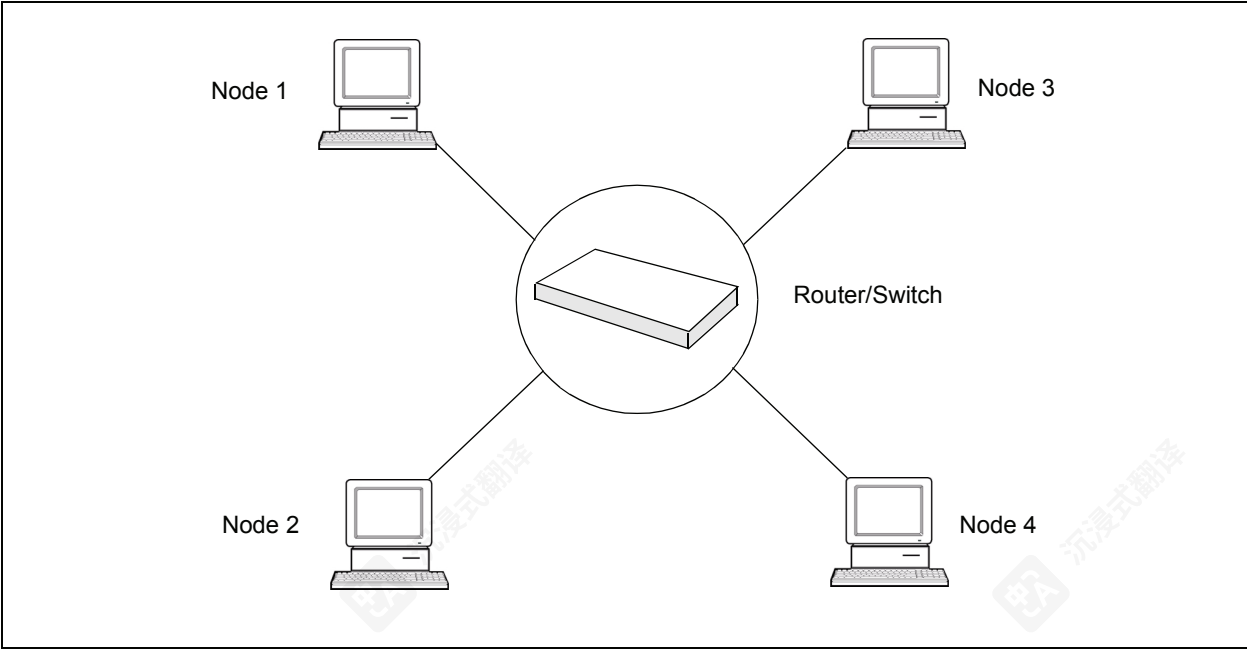


FIGURE 11: STAR TOPOLOGY (10Base-T, 100Base-TX)



全双工操作

虽然早期的以太网网络采用共享介质实现，并需要 CSMA/CD，但大多数现代以太网网络配置为点对点（图 10）或星型拓扑（图 11），可视为点对点连接的集合。

在任何配置中，由于每个节点最多仅连接一个其他节点，每个节点均可工作在全双工模式。在点对点/全双工配置下，不会发生冲突，因此无需使用 CSMA/CD 协议。每个节点可根据需要随时发送数据，只要满足数据包间隔对发送所施加的限制即可。

此外，介质的总吞吐量翻倍（即从 10 Mb/s 提升至 20 Mb/s，或从 100 Mb/s 提升至 200 Mb/s）。

全双工操作还具有消除因时隙时间而导致的网络直径限制的优势。

注意，并非所有介质类型都支持全双工。特别是以下类型不支持：

- 10Base2
- 10Base5
- 10Base-FP
- 10Base-FB
- 100Base-T4

图 10: 点对点拓扑学

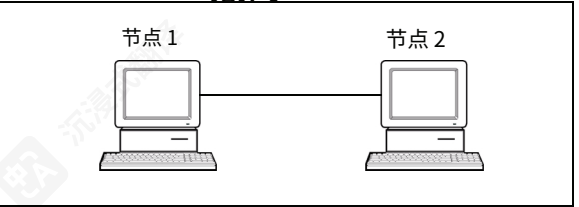
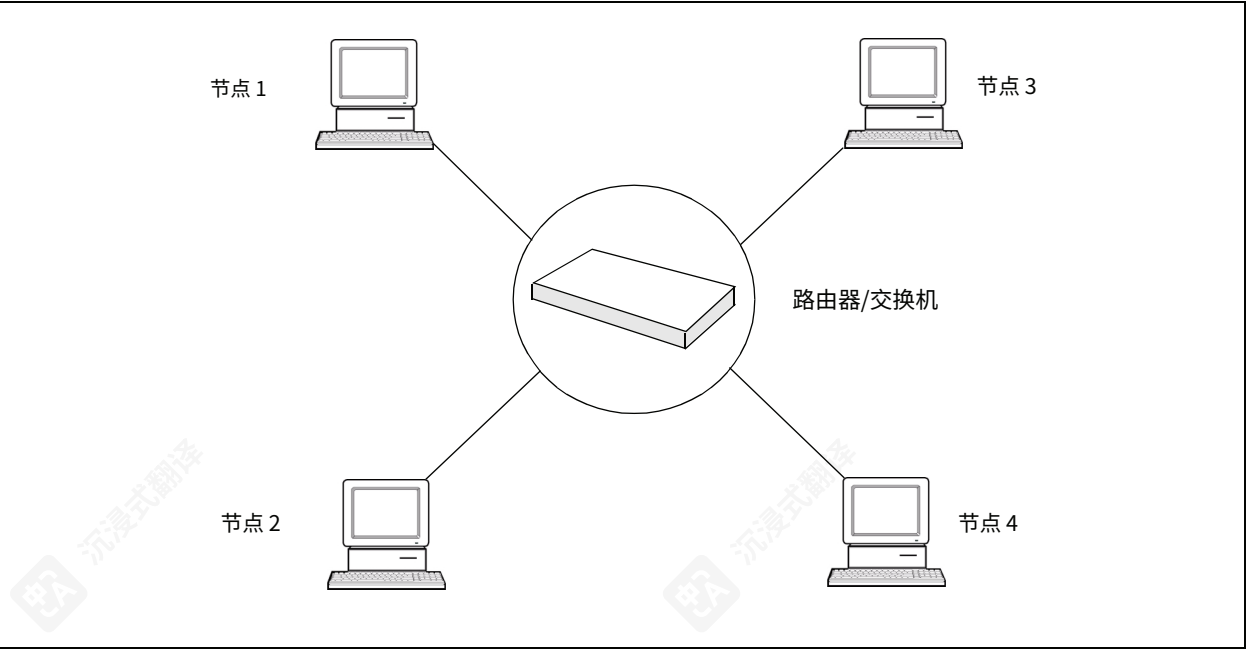


图 11: 星型拓扑 (10Base-T, 100Base-TX)



10 Mb/s STREAM CONTENTS

There are distinct differences between a 10 Mb/s and a 100 Mb/s stream, so let us discuss the contents and signaling of the 10 Mb/s stream first. This section describes how the frame shown in Figure 3 is actually transported over the physical medium (i.e., CAT5 cable, etc.).

The first step in transmission of a 10 Mb/s stream is to encode the data to be transmitted using Manchester encoding. Manchester encoding encodes a logical ‘0’ as a mid-bit low-to-high or high-to-low transition on the signal, and a logical ‘1’ as the opposite transition. In Ethernet, a logical ‘0’ is encoded as a high-to-low transition, while a logical ‘1’ is encode as a low-to-high transition. See Figure 12 for an example.

Manchester encoding is used because it provides high reliability and the ability to extract the clock from the data stream. However, it requires double the bandwidth of the data to be transmitted.

Since 10Base-T Ethernet uses differential signaling, the Manchester encoded signal is transmitted using differential signals, as shown in Figure 12.

The second step in transmission is to wave shape the output signals in order to meet the signal profiles defined in the IEEE 802.3 specification. These profiles are meant to ensure adequate signal propagation over the required lengths on the particular physical medium while minimizing unwanted EMI emissions.

Finally, the signal is transmitted over the cable using either voltage drive or current drive (which one depends on the particular IEEE 802.3 PHY) from an isolation transformer.

The differential voltage levels at the receiver are in the range of 350 mV to 3.1V.

10 Mb/s 流媒体内容

10 Mb/s 流与 100 Mb/s 流之间存在显著差异，因此我们先讨论 10 Mb/s 流的内容和信令。本节描述图 3 所示帧如何在物理介质（如 CAT5 电缆等）上进行传输。

传输 10 Mb/s 数据流的第一步是使用曼彻斯特编码对数据进行编码。曼彻斯特编码将逻辑 ‘0’ 编码为信号上的中位低到高或高到低跳变，将逻辑 ‘1’ 编码为相反的跳变。在以太网中，逻辑 ‘0’ 被编码为高到低跳变，而逻辑 ‘1’ 被编码为低到高跳变。参见图 12 了解示例。

曼彻斯特编码被采用，因为它提供了高可靠性以及从数据流中提取时钟的能力。然而，它需要传输数据的两倍带宽。

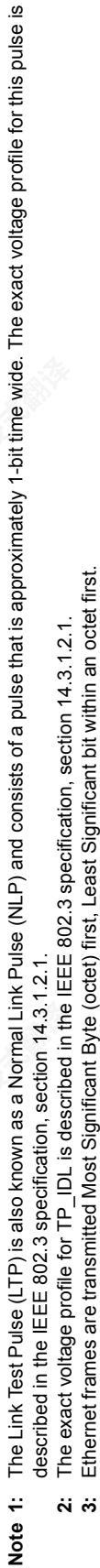
由于 10Base-T 以太网使用差分信号，曼彻斯特编码信号通过差分信号传输，如图 12 所示。

传输的第二步是对输出信号进行波形整形，以满足 IEEE 802.3 规范中定义的信号特性。这些特性旨在确保信号在特定物理介质上在所需长度内得到充分传播，同时最大限度地减少不需要的电磁干扰（EMI）发射。

最后，信号通过隔离变压器，采用电压驱动或电流驱动方式（具体取决于特定的 IEEE 802.3 物理层）在电缆上传输。

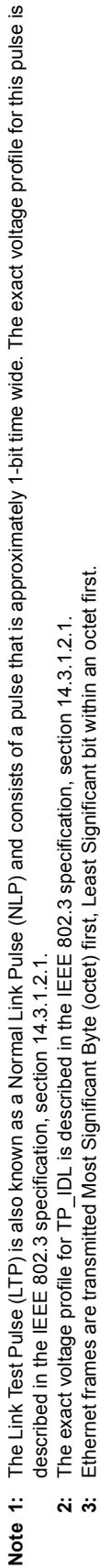
接收端处的差分电压电平范围为 350 mV 至 3.1V。

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- 1:** The Link Test Pulse (LTP) is also known as a Normal Link Pulse (NLP) and consists of a pulse that is approximately 100 ns wide and 100 mV high. The LTP is described in the IEEE 802.3 specification, section 14.3.1.2.1.
- 2:** The exact voltage profile for TP_IDLE is described in the IEEE 802.3 specification, section 14.3.1.2.1.
- 3:** Ethernet frames are transmitted Most Significant Byte (octet) first. Least Significant bit within an octet first.

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- 1:** The Link Test Pulse (LTP) is also known as a Normal Link Pulse (NLP) and consists of a pulse that is approximately 100 ns wide and 100 mV high. The LTP is described in the IEEE 802.3 specification, section 14.3.1.2.1.
- 2:** The exact voltage profile for TP_IDL is described in the IEEE 802.3 specification, section 14.3.1.2.1.
- 3:** Ethernet frames are transmitted Most Significant Byte (octet) first. Least Significant bit within an octet first.

100 Mb/s STREAM CONTENTS

Because Unshielded Twisted Pair (UTP) wires are low-pass in nature, the same encoding scheme that was used for 10Base-T will not work when we increase the speed by 10x, as is required for 100 Mb/s operation.

In addition, the power transmitted over certain types of physical links (i.e., phone lines, etc.) is limited to be less than approximately 30 MHz by regulatory guidelines. Therefore, a different encoding scheme is required for 100Base-T.

The encoding scheme used in 100Base-TX is known as Multi-Level Transition 3 (MLT3), and is shown in Figure 13. Each logical ‘0’ or ‘1’ is encoded as a transition to one of three levels (hence the ‘3’ in MLT3). The transition is always to the closest voltage level, and always in the same order (-1, 0, +1, 0, -1, ...). A logical ‘0’ is denoted by no transition, while a logical ‘1’ is denoted by a transition.

As an example, consider the bit sequence ‘11111’ shown in Figure 13. Since ‘1’ always equates to a transition, a constant sequence of ‘1’s will give us a transition on every bit, as shown in the figure.

By always transitioning to the closest voltage level, the transition times can always be minimized.

Because MLT-3 requires 4 transitions (-1 to 0 to +1 to 0 to -1) to complete a full cycle, the maximum fundamental frequency is reduced by 4, from 125 MHz to 31.25 MHz. This meets our requirement for power transmission at no higher than approximately 30 MHz.

The non-encoded signal frequency spectrum is 125 MHz, instead of the expected 100 MHz, because of 4B/5B encoding, which is discussed in the next section.

4B/5B Encoding

In addition to the physical encoding of MLT3, 100Base-TX introduces a logical encoding called 4B/5B, or sometimes “Block Coding”. There are two primary requirements that 100Base-TX encoding must meet.

First, it must solve the problem of clock recovery in long streams of transmitted ‘0’s. In MLT3, as you recall, a ‘0’ is denoted by the lack of a transition in the transmitted signal. With no explicit clock, the transmit and receive nodes would soon become out of synchronization due to various jitter introducing effects. This would eventually result in the corruption of data.

Secondly, it must allow for transmission of not only data, but also of signaling codes, such as Start-of-Stream, End-of-Stream, Error and Idle.

The solution to these problems that 100 Mb/s Ethernet implements is to encode each 4 bits of data into 5 bits on the transmission medium. The translation from 4 bits to 5 bits is shown in Table 3. This means the actual transmission rate over the physical medium for 100 Mb/s Ethernet is 125 Mb/s.

If we look closely at the coding for all of the codes (except /H/, which is an error code), we will see the actual transmitted value always contains at least two ‘1’s, which will result in a minimum of two transitions in the MLT3 waveform for any data transmitted. This addresses the issue of clock recovery.

With 2⁵ encodings for 16 data values, we now have 16 extra values that can be used to transmit signaling data. These include the following:

- Idle, which replaces the Normal Link Pulses (NLPs) used in 10Base-T
- Start-of-Stream Delimiter (SSD), which replaces the first octet of the Preamble in 10Base-T
- End-of-Stream Delimiter (ESD), which replaces the TP_IDL waveform used in 10Base-T
- Transmit error, which has no equivalent in 10Base-T

TABLE 3: 4B/5B ENCODING

Code	Value	Definition
0	11110	Data 0
1	01001	Data 1
2	10100	Data 2
3	10101	Data 3
4	01010	Data 4
5	01011	Data 5
6	01110	Data 6
7	01111	Data 7
8	10010	Data 8
9	10011	Data 9
A	10110	Data A
B	10111	Data B
C	11010	Data C
D	11011	Data D
E	11100	Data E
F	11101	Data F
I	11111	Idle
J	11000	SSD (Part 1)
K	10001	SSD (Part 2)
T	01101	ESD (Part 1)
R	00111	ESD (Part 2)
H	00100	Transmit Error

100 Mb/s 流媒体内容

由于非屏蔽双绞线（UTP）具有低通特性，当需要将速度提高 10 倍以满足 100 Mb/s 运行需求时，用于 10Base-T 的相同编码方案将无法工作。

此外，某些类型的物理链路（如电话线等）传输的功率受监管指南限制，频率低于约 30 MHz。因此，100Base-T 需要采用不同的编码方案。

100Base-TX 中使用的编码方案称为多级跳变 3（MLT3），如图 13 所示。每个逻辑 ‘0’ 或 ‘1’ 被编码为向三个电平之一（因此 MLT3 中的 ‘3’）的跳变。跳变总是指向最近的电压电平，并且总是按相同顺序进行（-1, 0, +1, 0, -1, ...）。逻辑 ‘0’ 由无跳变表示，而逻辑 ‘1’ 由跳变表示。

例如，考虑图 13 中所示的位序列 ‘11111’。由于 ‘1’ 始终表示转换，因此 ‘1’ 的恒定序列将在每一位上产生转换，如图所示。

通过始终切换到最近的电压等级，可以始终最小化转换时间。

由于 MLT-3 需要 4 次转换（-1 到 0 到 +1 到 0 到 -1）才能完成一个完整周期，因此最大基频降低了 4，从 125 MHz 降至 31.25 MHz。这满足了我们在不超过约 30 MHz 的情况下进行功率传输的要求。

未编码的信号频谱为 125 MHz，而非预期的 100 MHz，这是由于 4B/5B 编码所致，将在下一节中讨论。

4B/5B 编码

除了 MLT3 的物理编码外，100 4 5 Base-TX 还引入了称为 B/ B 的逻辑编码，或有 100Base-TX 编码必须满足两个主要要求。

首先，必须解决在长串传输的 ‘0’ 中时钟恢复的问题。在 MLT3 中，正如你所知，‘0’ 由传输信号中无跳变来表示。由于没有显式时钟，收发节点会因各种抖动引入效应而很快失去同步。这最终会导致数据损坏。

其次，它必须允许传输数据，以及诸如流开始、流结束、错误和空闲等信令码。

100 Mb/s 以太网解决这些问题的方法是：将每个 4 位数据编码为传输介质上的 5 位。4 位到 5 位的转换如表 3 所示。这意味着 100 Mb/s 以太网在物理介质上的实际传输速率为 125 Mb/s。

如果我们仔细查看所有代码（/H/ 除外，它是错误代码）的编码，会发现实际传输的值始终至少包含两个 ‘1’，这将导致任何传输的数据在 MLT3 波形中至少产生两个跳变。这解决了时钟恢复的问题。

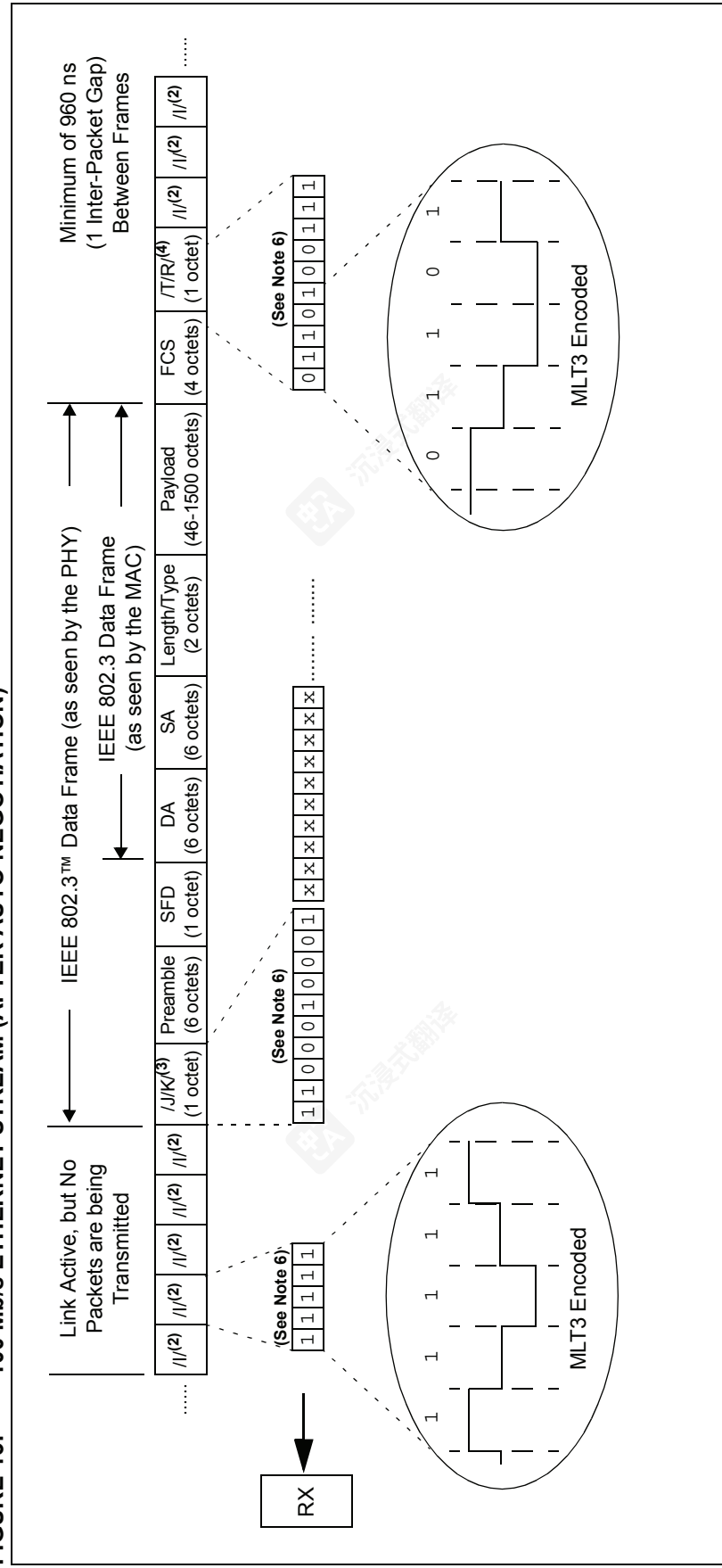
拥有 16 个数据值的 25 种编码，我们现在拥有了 16 个额外的值，可用于传输信令数据。其中包括以下内容：

- 空闲，用于替代 10Base-T 中使用的正常链路脉冲（NLPs）
- 流开始分隔符 (SSD)，它替换 10Base-T 中前导序列的第一个字节
- 流结束符 (ESD)，它替换 10Base-T_ 中使用的 TP IDL 波形
- 传输错误，在 10Base-T 中没有等效项

TABLE 3: 4B/5B 编码

Code	值	定义
0	11110	数据 0
1	01001	数据 1
2	10100	数据 2
3	10101	数据 3
4	01010	数据 4
5	01011	数据 5
6	01110	数据 6
7	01111	数据 7
8	10010	数据 8
9	10011	数据 9
A	10110	数据 A
B	10111	数据 B
C	11010	数据 C
D	11011	数据 D
E	11100	数据 E
F	11101	数据 F
I	11111	Idle
J	11000	固态硬盘 (第 1 部分)
K	10001	固态硬盘 (第二部分)
T	01101	ESD（第一部分）
R	00111	静电放电（第 2 部分）
H	00100	传输错误

FIGURE 13: 100 Mb/s ETHERNET STREAM (AFTER AUTO-NEGOTIATION)^(1,5)



Note 1: Due to 4B/5B encoding, each octet seen by the PHY/MAC corresponds to 10 bits on the physical medium. Thus, the data rate on the physical medium is actually 125 Mb/s.

2: `// = Idle code group ('11111')`.

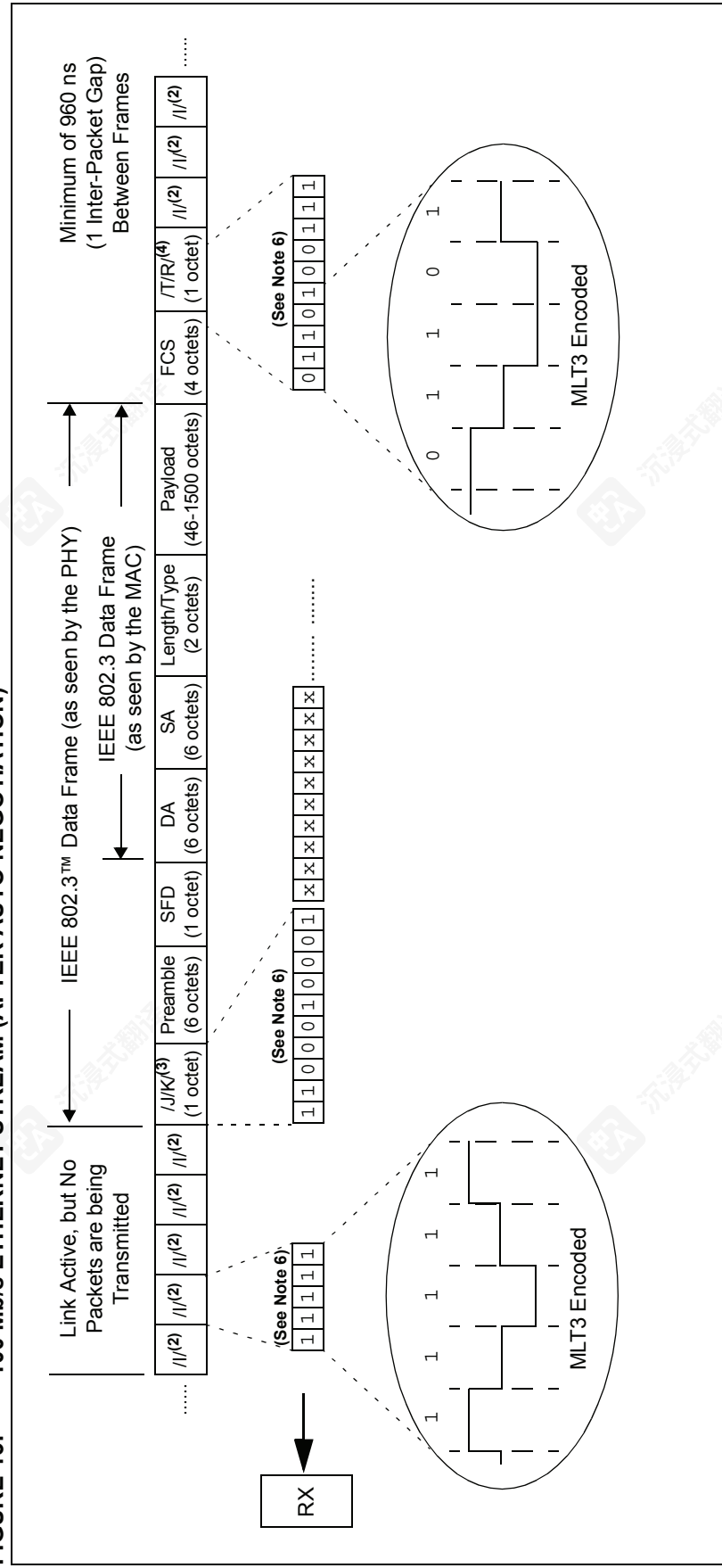
3: /J/K/ = Start-of-Stream Delimiter (SSD) code group ("10001 11000").

4: /T/R/ = End-of-Stream Delimiter (ESD) code group ('00111 01101').

5: Ethernet frames are transmitted Most Significant Byte (octet) first, Least Significant bit within an octet first.

6: Unlike 10 Mb/s Ethernet, signals on the physical medium are scrambled to reduce radiated emissions. The bit streams shown are unscrambled and are shown for the sole purpose of understanding how MLT3 encoding works.

FIGURE 13: 100 Mb/s ETHERNET STREAM (AFTER AUTO-NEGOTIATION)^(1,5)



Note 1: Due to 4B/5B encoding, each octet seen by the PHY/IMAC corresponds to 10 bits on the physical medium. Thus, the data rate on the physical medium is actually 125 Mb/s.

2: `// = Idle code group ('11111')`.

3: /J/K/ = Start-of-Stream Delimiter (SSD) code group ("10001 11000").

4: $\text{TR}/ = \text{End-of-Stream Delimiter (ESD) code group ('00111 01101')}$.

5: Ethernet frames are transmitted Most Significant Byte (octet) first, Least Significant bit within an octet first.

5: Unlike 10 Mb/s Ethernet, signals on the physical medium are scrambled to reduced radiated emissions. The bit streams shown are unscrambled and are shown for the sole purpose of understanding how MLT3 encoding works.

ENCODING/DECODING OVERVIEW

So far, we have discussed the encoding and decoding of 10Base-T, where Manchester encoding is used, and of 100Base-T, where NRZI, MLT3 and 4B/5B encoding are all used.

With Manchester encoding, where a single transition represents a single bit, the 10 Mb/s data rate translates into a 10 MHz bandwidth requirement on the medium. Noise immunity is added through the use of differential signalling on the medium.

How then, do all of the encoding methods employed on 100Base-TX combine to produce a final signal to be transmitted over the medium?

Figure 14 shows a simplified block diagram of a 100Base-TX PHY, with the bandwidth requirements at each stage. From this diagram, we can see that even though the effective data rate of the stream is increased to 125 Mb/s due to 4B/5B encoding, the required bandwidth of the physical medium is actually much smaller than 125 MHz.

编码/解码概览

到目前为止，我们讨论了 10Base-T 的编码和解码，其中使用曼彻斯特编码，以及 100Base-T 的编码和解码，其中使用了 NRZI、MLT3 和 4B/5B 编码。

采用曼彻斯特编码，单个跳变代表一个比特，10 Mb/s 的数据速率对应于介质上 10 MHz 的带宽需求。通过在介质上使用差分信号来增强抗噪声能力。

那么，100Base-TX 中采用的所有编码方法如何协同工作，以产生最终用于在介质上传输的信号？

图 14 展示了 100Base-TX 物理层的简化框图及各阶段的带宽需求。从该图可以看出，尽管由于 4B/5B 编码，流的有效数据速率提升至 125 Mb/s，但物理介质的实际所需带宽却远小于 125 MHz。

FIGURE 14: SIMPLIFIED 100Base-TX PHY BLOCK DIAGRAM

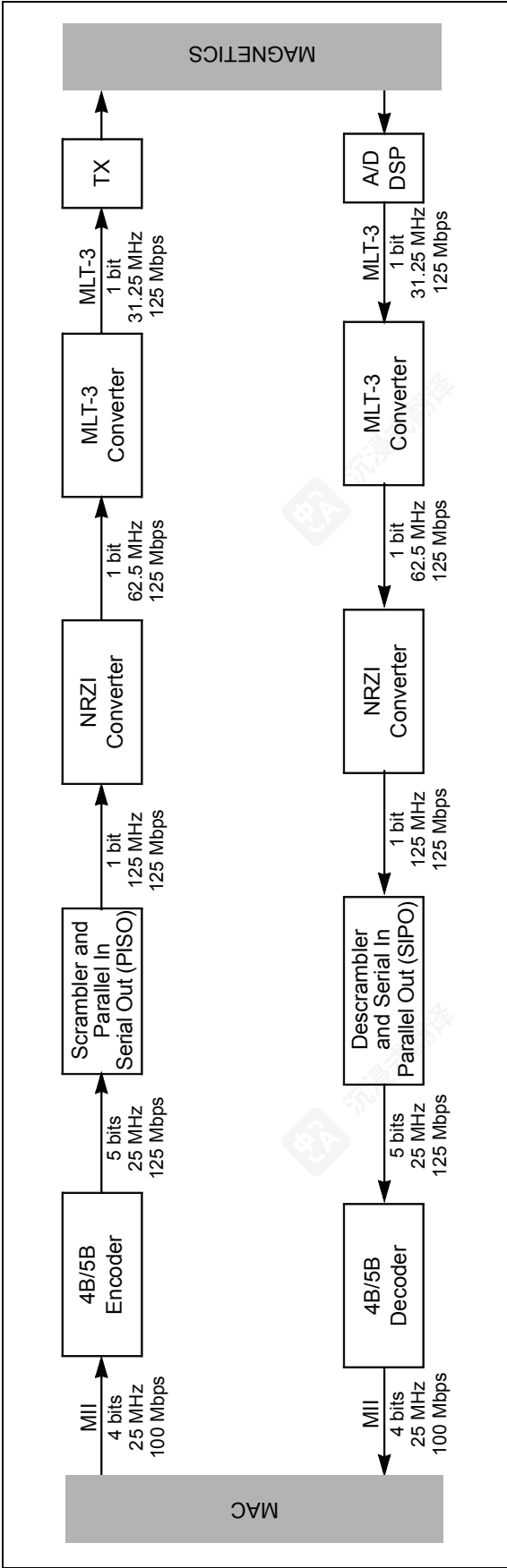
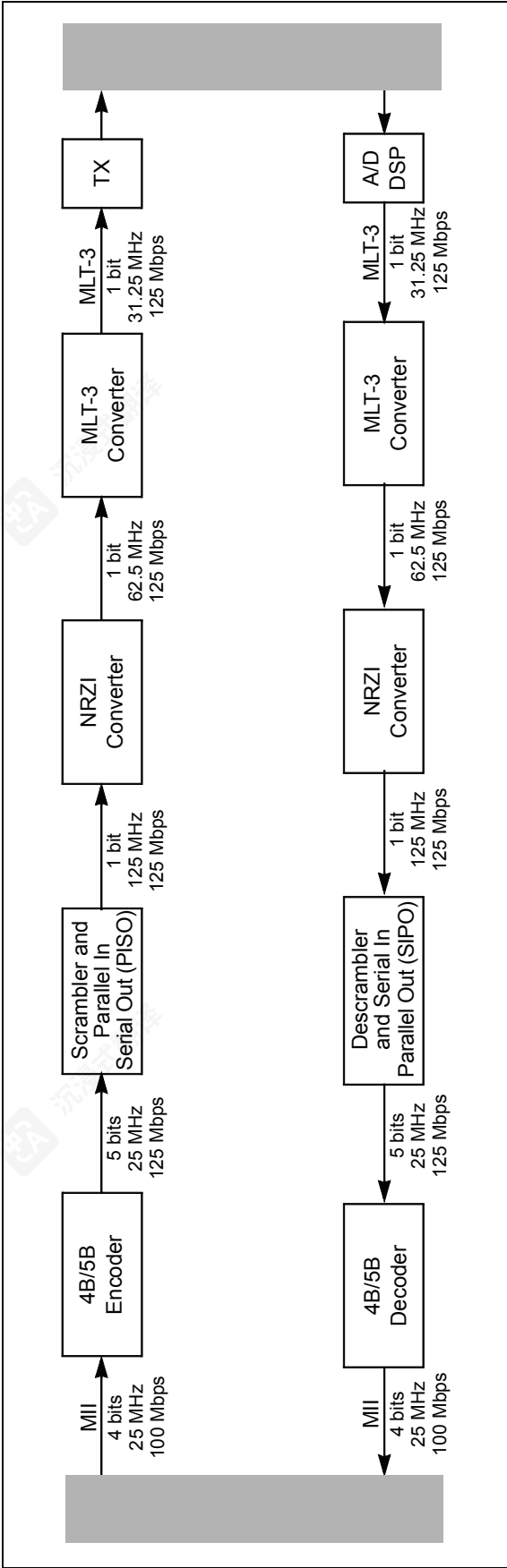


FIGURE 14: SIMPLIFIED 100Base-TX PHY BLOCK DIAGRAM



AUTO-NEGOTIATION

Auto-negotiation is the process by which two nodes communicate their respective abilities (speed, duplex, support for pause frames, etc.) in order to choose the highest common ability for both ends of the link. Auto-negotiation takes place at link initialization, and is backward compatible (i.e., does not break nodes that do not support auto-negotiation). Auto-negotiation is optional for 10Base-T and 100Base-T, but required for gigabit Ethernet.

Auto-negotiation is performed through the use of Fast Link Pulses (FLPs) shown in Figure 15. FLPs are similar to Normal Link Pulses (NLPs), but are transmitted in a burst of 17-33 pulses (called a link code word) between NLPs. Given the minimum inter-space timing of about 62.5 μs for FLPs, and the bits times of 100 ns (10 Mb/s) and 10 ns (100 Mb/s), it should be clear that FLPs are not interpreted as valid data by Ethernet nodes. In fact, FLPs are interpreted by nodes that do not support auto-negotiation as NLPs and are ignored. Nodes that support auto-negotiation, but do not receive any FLPs from the opposite end of the link automatically default to the lowest common ability (typically half-duplex 10Base-T) by default. In addition, some Ethernet PHYs have the capability to distinguish between 10 Mb/s and 100 Mb/s operation (based on the physical encoding seen on the link), a feature known as parallel detection. Of course, it is still possible to configure each end of the link manually to settle on a common ability, but this must be done in software.

It is possible that two nodes both support auto-negotiation, but no common ability is found. In this case, no link is established.

The maximum number of link code words sent by each node is not defined by the IEEE 802.3, but each node that supports auto-negotiation must be capable of transmitting an auto-negotiation base page.

The 16-bit (1 link code word) base page defines the abilities of the node, and is defined as follows:

- Bits 0-4:

Known as the selector field (S0-S4), this field defines the type of LAN technology used. For IEEE 802.3 Ethernet, this field is set to ‘10000’.
- Bits 5-12:

Known as the technology ability field (A0-A7), this field defines the capability of the node.
- Bits 5-9:

Defines the link type, with the following defined priority (in order of highest priority to the lowest priority):

100Base-TX full-duplex (bit 3 set)

100Base-T4 (bit 4 set)

100Base-TX (bit 2 set)

10Base-T full-duplex (bit 1 set)

10Base-T (bit 0 set)
- Bit 10:

0 = Pause not enabled
1 = Pause enabled
- Bit 11:

Supports asymmetric pause operation for full-duplex links
- Bit 12:

Extended next pages bit, used only with gigabit Ethernet nodes.
- Bit 13:

Known as the Remote Fault (RF) indicator bit, this bit indicates a remote Fault.
- Bit 14:

Known as the Acknowledge (Ack) bit, this mandatory bit signals the receipt of an FLP message. An FLP message must be received identically three consecutive times before it is considered correct and Acknowledged.
- Bit 15:

Known as the Next Page (NP) bit, this bit indicates whether a next page link code word is following the base page. Next page words are used to transmit extra information between linked nodes during auto-negotiation, and is an optional capability.

自动协商

自动协商是两端节点相互通信以交换各自能力（如速率、双工模式、暂停帧支持等）的过程，从而为链路两端选择共同支持的最高能力。自动协商在链路初始化时进行，具有向后兼容性（即不会破坏不支持自动协商的节点）。自动协商在 10Base-T 和 100Base-T 中是可选的，但在千兆以太网中是必需的。

自动协商通过图 15 所示的快速链路脉冲（FLPs）执行。FLPs 类似于普通链路脉冲（NLPs），但在 NLPs 之间以 17-33 个脉冲的突发形式传输（称为链路码字）。鉴于 FLPs 的最小间隔时间约为 62.5 μs，而 100 ns（10 Mb/s）和 10 ns（100 Mb/s）的位时间，以太网节点显然不会将 FLPs 解释为有效数据。事实上，不支持自动协商的节点将 FLPs 解释为 NLPs 并予以忽略。支持自动协商但未从链路对端接收任何 FLPs 的节点会自动默认到最低共同能力（通常为半双工 10Base-T）。此外，某些以太网物理层具有区分 10 Mb/s 和 100 Mb/s 操作的能力（基于链路上看到的物理编码），该功能称为并行检测。当然，仍然可以手动配置链路的每一端以达成共同能力，但这必须在软件中完成。

可能两个节点都支持自动协商，但未找到共同能力。在这种情况下，不会建立连接。

每个节点发送的链路代码字最大数量由 IEEE 802.3 未定义，但支持自动协商的每个节点必须能够传输自动协商基础页。

16 位（1 个链接码字）基页定义了节点的权限，定义如下：

- 位 0-4:

被称为选择器字段（S0-S4），该字段定义了所使用的 LAN 技术类型。对于 IEEE 802.3 以太网，该字段设置为 ‘10000’。
- 位 5-12:

称为技术能力字段（A0-A7），该字段定义节点的 capability。
- 位 5-9:

定义链接类型，具有以下定义的优先级（按从高到低排序）：

100Base-TX 全双工模式 (第 3 位)

100Base-T4（第 4 位为 1)

100Base-TX（第 2 位置 1)

10Base-T 全双工（第 1 位置 1)

10Base-T（第 0 位置 1)
- 位 10:

0 =暂停未启用
1 = 暂停已启用
- 位 11:

支持全双工链路的非对称暂停操作
- 位 12:

扩展下一页位，仅与千兆以太网节点配合使用。
- 位 13:

被称为远程故障（RF）指示器 bit, 此位指示远程故障
- 位 14:

被称为 Acknowledge（Ack）位，这是一个必需位，用于指示收到 FLP 消息。只有当 FLP 消息被连续三次以相同形式收到时，才被视为正确并被确认。
- 位 15:

该位被称为“下一页（NP）”位，用于指示下一页链接码字是否跟在基础页之后。下一页字用于在自动协商期间于链接节点之间传输额外信息，这是一种可选功能。

FIGURE 15: FAST LINK PULSES

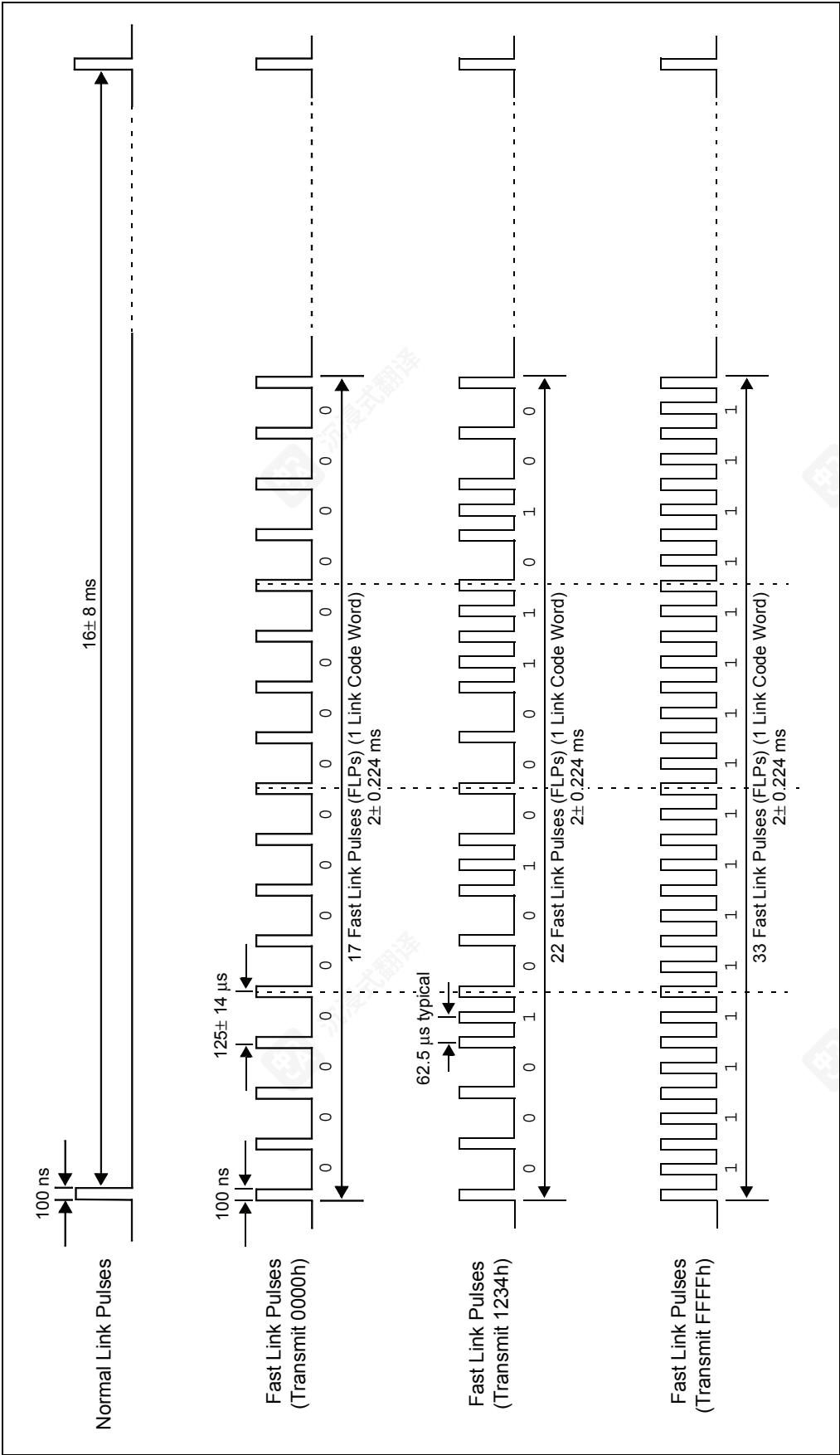
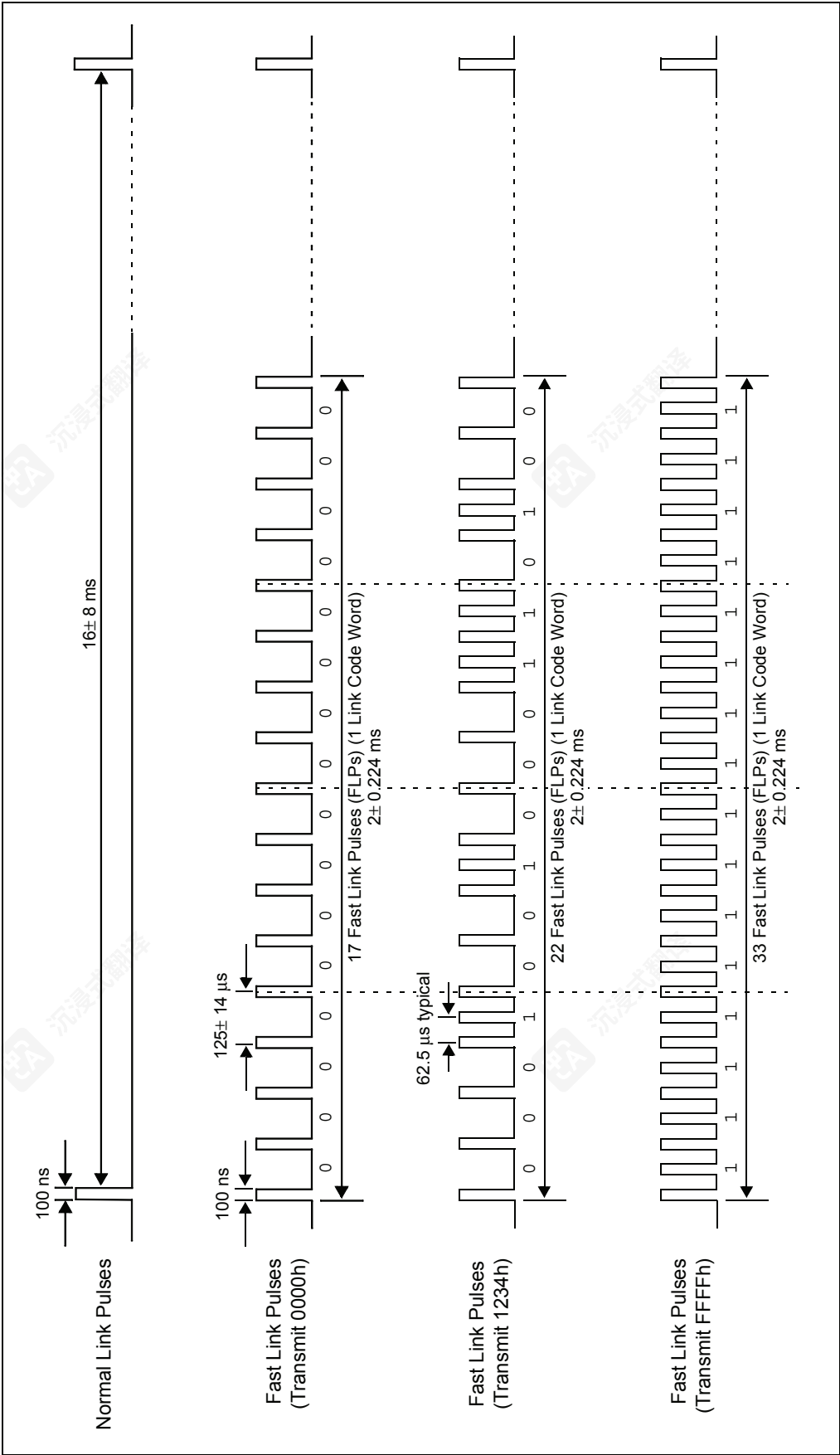


FIGURE 15: FAST LINK PULSES



AUTO-CROSSOVER

In a properly configured Ethernet connection, the TX port of one node is connected to the RX port of the other node, and vice-versa.

In star topology UTP Ethernet networks, this crossover is typically done in the switch/hub/router's connection to the Ethernet jack. As a result, most UTP Ethernet cables have a 1-to-1 pin mapping between the connectors on the ends of the cable. Cables of this type are commonly referred to as “straight-through cables”.

However, a different type of cable exists, called a “crossover cable”. This type of cable internally crosses the TX and RX port on one end of the cable to the RX and TX port on the other end of the cable, respectively. This type of cable allows two end Ethernet devices to communicate with each other when directly connected as a point-to-point network. Additionally, crossover cables allow a switch/hub/router to communicate with another switch/hub/router. Using an incorrect cable type will not damage compliant Ethernet nodes, but neither node will be able to communicate or detect a link.

To eliminate cabling mismatches and reduce consumer frustration, a feature called auto-crossover may optionally be implemented in a node. When implemented, an auto-crossover capable node will automatically swap its TX/RX pins between TX and RX until a link is established. In this manner, either a crossover or patch cable may be used with the node with the same results. It is only necessary that one node in a linked pair implement auto-crossover. Most modern switches, routers, etc., implement auto-crossover.

Note that this functionality is different from “auto-polarity”, where a node may automatically switch between positive and negative signals on a TX port or on an RX port. The two functions serve different purposes and are unrelated.

Auto-crossover is also sometimes referred to as Auto-MDIX, due to the fact that the crossover (“X” in Auto-MDIX) occurs at the MDI layer in the node (see Figure 6).

自动切换

在正确配置的以太网连接中，一个节点的 TX 端口连接到另一个节点的 RX 端口，反之亦然。

在星型拓扑 UTP 以太网网络中，这种交叉连接通常在交换机/集线器/路由器'与以太网接口的连接中进行。因此，大多数 UTP 以太网电缆在电缆两端的连接器之间具有 1 对 1 的引脚映射。此类电缆通常被称为“直通电缆”。

然而，存在另一种类型的电缆，称为“交叉电缆”。这种电缆在电缆的一端将 TX 和 RX 端口内部交叉连接到电缆另一端的 RX 和 TX 端口。这种电缆允许两个直接连接的点对点网络中的以太网端点设备相互通信。此外，交叉电缆允许交换机/集线器/路由器与另一台交换机/集线器/路由器通信。使用错误的电缆类型不会损坏符合标准的以太网节点，但两个节点都无法通信或检测链路。

为消除线缆不匹配问题并减少用户困扰，可在节点中可选地实现自动交叉功能。启用该功能后，具备自动交叉能力的节点将在链路建立前自动交换其 TX/RX 引脚，直至链路成功建立。通过这种方式，无论使用交叉线还是直通线连接该节点，结果均相同。只需在成对连接的节点中有一个实现自动交叉功能即可。大多数现代交换机、路由器等设备均支持自动交叉功能。

请注意，此功能与“auto-polarity”不同，后者允许节点在 TX 端口或 RX 端口上自动在正信号和负信号之间切换。这两个功能用途不同且互不相关。

自动交叉有时也称为自动 MDIX，因为交叉（“X”在自动 MDIX 中）发生在节点的 MDI 层（见图 6）。

REFERENCES

The following documents are referenced in this application note:

- IEEE 802.3 Specification
- Associated IEEE Supplements (see Table 4)

TABLE 4: MOST COMMON SPECIFICATION SUPPLEMENTS

Supplement	Year	Description
IEEE 802.3a	1985	10Base-2 Thin Ethernet
IEEE 802.3c	1985	10 Mb/s Repeater Specification
IEEE 802.3d	1987	Fiber Optic Inter-Repeater Link
IEEE 802.3i	1990	10Base-T Twisted Pair
IEEE 802.3j	1993	10Base-F Fiber Optic
IEEE 802.3u	1995	100Base-T Fast Ethernet and Auto-Negotiation
IEEE 802.3x	1997	Full-Duplex Standard
IEEE 802.3z	1998	1000Base-X Gigabit Ethernet (SX, LX, CX)
IEEE 802.3ab	1999	1000Base-T Gigabit Ethernet over Twisted Pair
IEEE 802.3ac	1998	Frame Size Extension to 1522 Octets for VLAN Tagging
IEEE 802.3ad	2000	Link Aggregation for Parallel Links
IEEE 802.3af	2003	Power Over Ethernet (PoE)

参考文献

本文档中引用的文档如下：

- IEEE 802.3 规范
- 关联的 IEEE 增刊（见表 4）

表 4：最常见的规格补充

补充	Year	描述
IEEE 802.3a	1985	10Base-2 细缆以太网
IEEE 802.3c 标准	1985	10 Mb/s 中继器规格
IEEE 802.3d	1987	光纤中继器链路
IEEE 802.3i	1990	10Base-T 双绞线
IEEE 802.3j	1993	10Base-F 光纤
IEEE 802.3u	1995	100Base-T 快速以太网和自动协商
IEEE 802.3x	1997	全双工标准
IEEE 802.3z	1998	1000Base-X 千兆以太网（SX、LX、CX）
IEEE 802.3ab	1999	1000Base-T 双绞线千兆以太网
IEEE 802.3ac	1998	帧大小扩展至 1522 字节用于 VLAN 标记
IEEE 802.3ad	2000	链路聚合用于并行链路
IEEE 802.3af	2003	以太网供电 (PoE)

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NOTES:

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